COURSE SLIDES · WSQ

QUANTUM

Securing the Future with Quantum Computing and Cryptography

WSQ Course Code: TGS-2024043419

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 13.0 · 11 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1. Trainer displays the digital attendance QR code from the SSG portal
2. Scan the QR code with your mobile phone camera
3. Submit your attendance — AM, PM and Assessment, every day
4. 75% minimum attendance is required for assessment and funding

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS
PhD; ACTA / DACE certified adult educator.

EXPERTISE
Quantum computing, quantum cryptography, data science and AI engineering.

DELIVERS
WSQ courses on quantum computing, cybersecurity and applied machine learning.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your background in physics, mathematics or programming (if any).
- What you want to be able to do with quantum computing after this course.

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HOUSEKEEPING

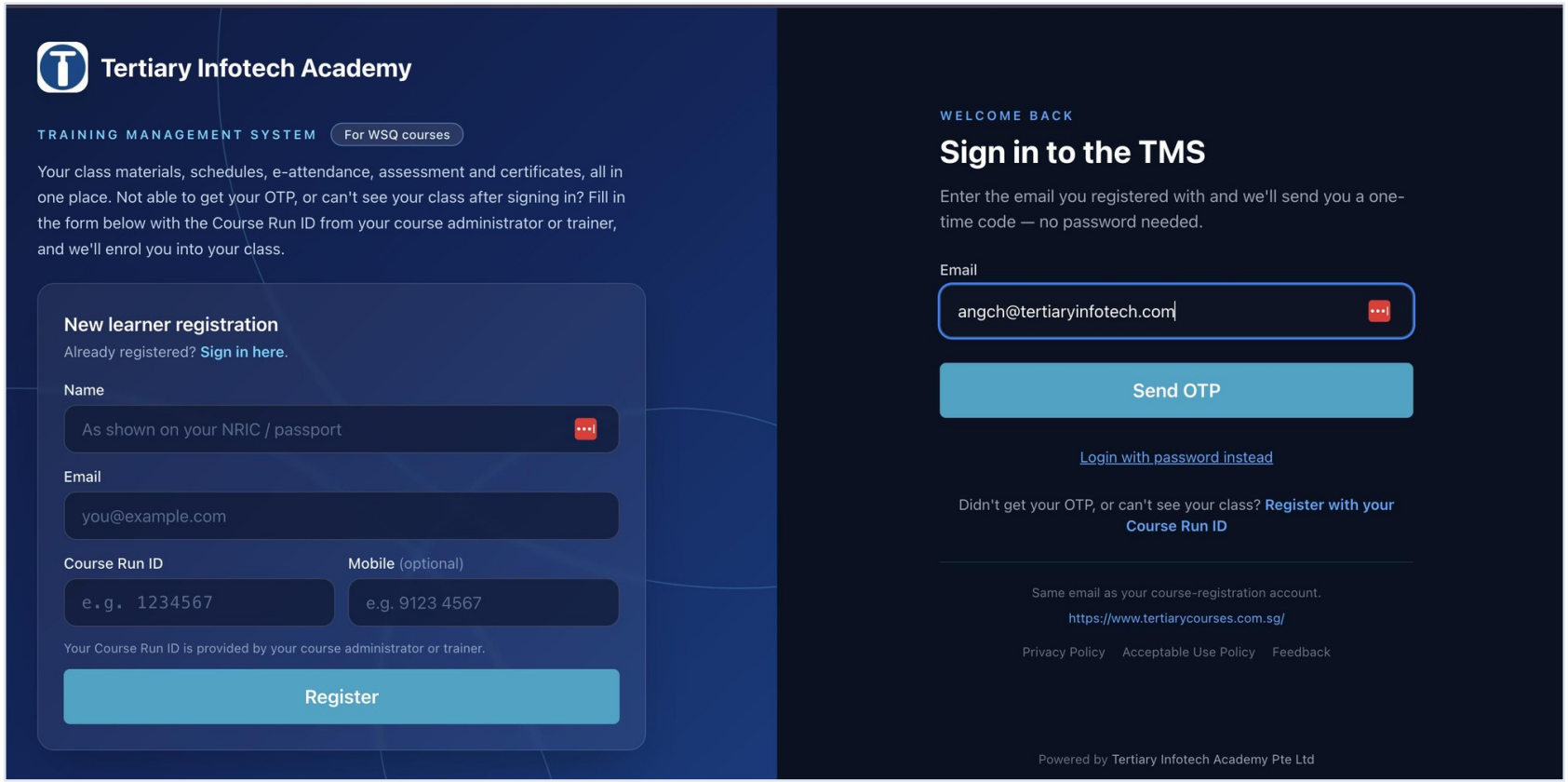
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding.

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COURSE PORTAL

Login to LMS and Download Course Material

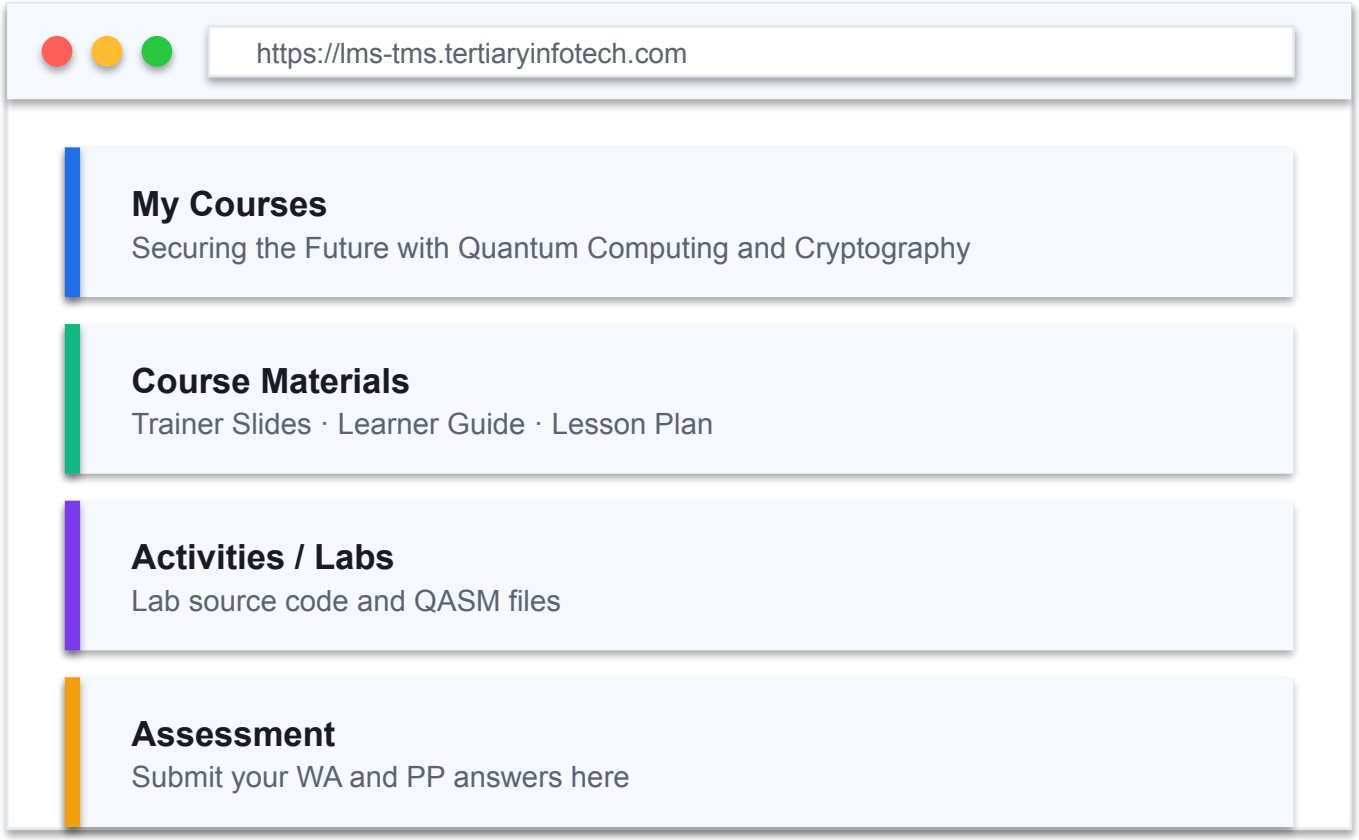


Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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COURSE PORTAL · LMS/TMS

Download Your Course Material



1. Go to lms-tms.tertiaryinfotech.com and sign in with the account you were issued.
2. Open My Courses and select this course.
3. Download the Trainer Slides, Learner Guide and Lesson Plan.
4. Keep them open — the assessment is OPEN BOOK.

Download the slides and Learner Guide BEFORE the assessment — they are your permitted open-book materials.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Mathematical models

Design the mathematical models required for quantum computing applications.

2

LO2 · Physical methodologies

Decide on appropriate physical methodologies for quantum computing problems.

3

LO3 · Two-level models

Direct two-level quantum models for quantum computing.

4

LO4 · Business applicability

Evaluate quantum computing applicability in the business context.

5

LO5 · Quantum gates

Spearhead the application of quantum gates to quantum computing domains.

6

LO6 · Algorithm selection

Establish guidelines for selecting quantum computing algorithms.

7

LO7 · Synthesis

Synthesize quantum methodologies to address complex quantum computing problems.

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SCHEDULE

Lesson Plan — 4 Days, 8 hours/day

Days 1–2

- Day 1 — Quantum Physics Fundamentals & Quantum Computing Systems
  - Topic 1: Quantum Physics Fundamentals (Activity 1)
  - Topic 2: Quantum Computing Systems (Activities 2–3)
- Day 2 — Quantum Gates, Circuits and Quantum Protocols
  - Topic 3: Quantum Gates and Circuits (Activities 4–5)
  - Topic 4: Quantum Protocols (Activities 6–8)

Days 3–4

- Day 3 — Quantum Algorithms and Quantum Cryptography
  - Topic 5: Quantum Algorithms (Activities 9–12)
  - Topic 6: Quantum Cryptography & Safety (Activities 13–14)
- Day 4 — Applications, Error Correction and Assessment
  - Topic 7: Applications (Activities 15–16)
  - Topic 8: Quantum Error Correction (Activities 17–18)
  - Final Assessment: WA (1 hr) + PP (1 hr)

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BEFORE YOU BEGIN

Briefing for Assessment

1

Phones away

Place phones and other materials under the table or on the floor.

2

No recording

No photos or recording of assessment scripts.

3

No discussion

Work individually — no discussion during the assessment.

4

Black or blue pen

Use a black/blue pen for hard-copy assessments.

5

No correction fluid

No liquid paper or correction tape on any script.

6

Time is called

Scripts are collected when the time is up.

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FINAL ASSESSMENT

Assessment

Written Assessment (WA)

- Short-Answer Questions (SAQ)
- 1 hour · open book
- Tests underpinning KNOWLEDGE
- Covers all eight topics

Practical Performance (PP)

- Hands-on quantum tasks
- 1 hour · open book
- Tests PRACTICAL ability
- Built on the class activities

OPEN BOOK — course slides, Learner Guide and approved materials only

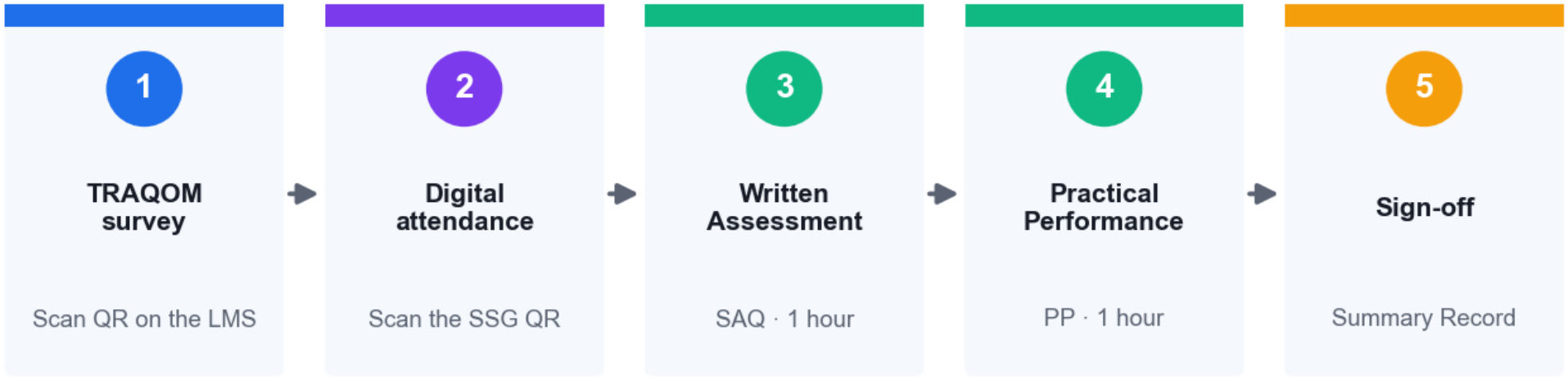
Eligibility: minimum 75% attendance + assessed Competent. An appeal process is available.

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ON ASSESSMENT DAY

Assessment Flow

What happens on assessment day



Open book — course slides, Learner Guide and approved materials only

Complete every step in order — the TRAQOM survey and digital attendance are mandatory for funding.

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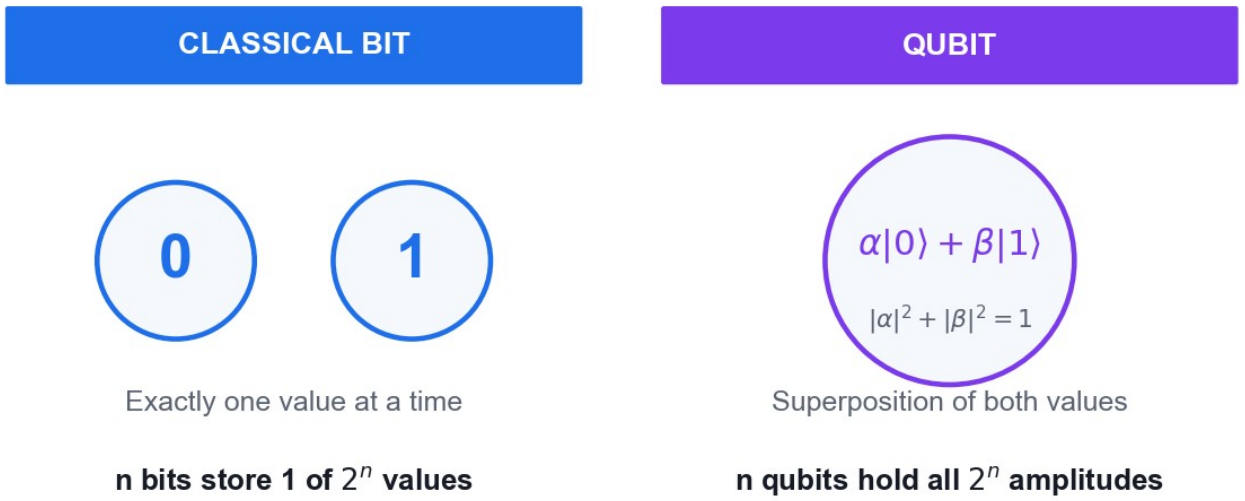
GETTING STARTED

Why Quantum, Why Now

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THE CORE IDEA

What Makes a Qubit Different



Classical bit

Holds exactly one value — 0 or 1 — at any instant.

Qubit

Holds a superposition of both, described by two complex amplitudes.

Measurement

Collapses the superposition to a single classical outcome.

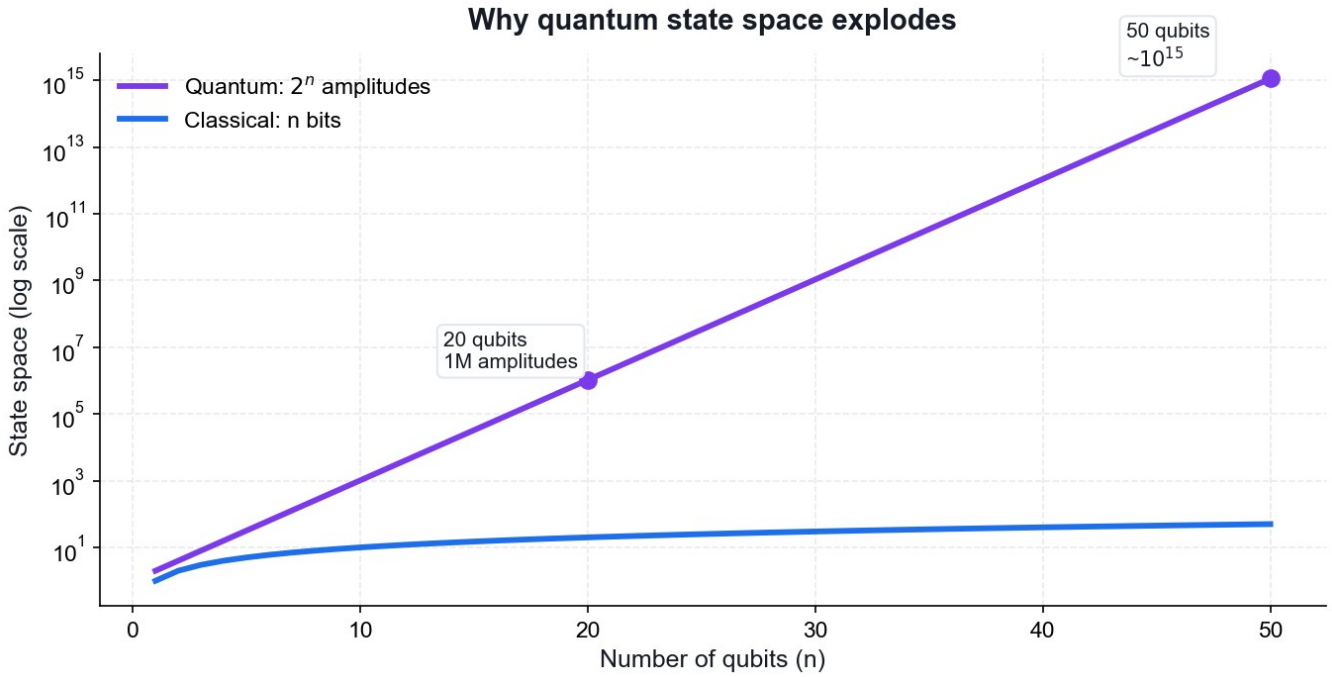
The payoff

n qubits carry 2^n amplitudes at once — quantum parallelism.

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WHY IT MATTERS

The State Space Explosion



50 qubits

~10^{15} amplitudes — beyond any classical supercomputer's memory.

300 qubits

More amplitudes than atoms in the observable universe.

But

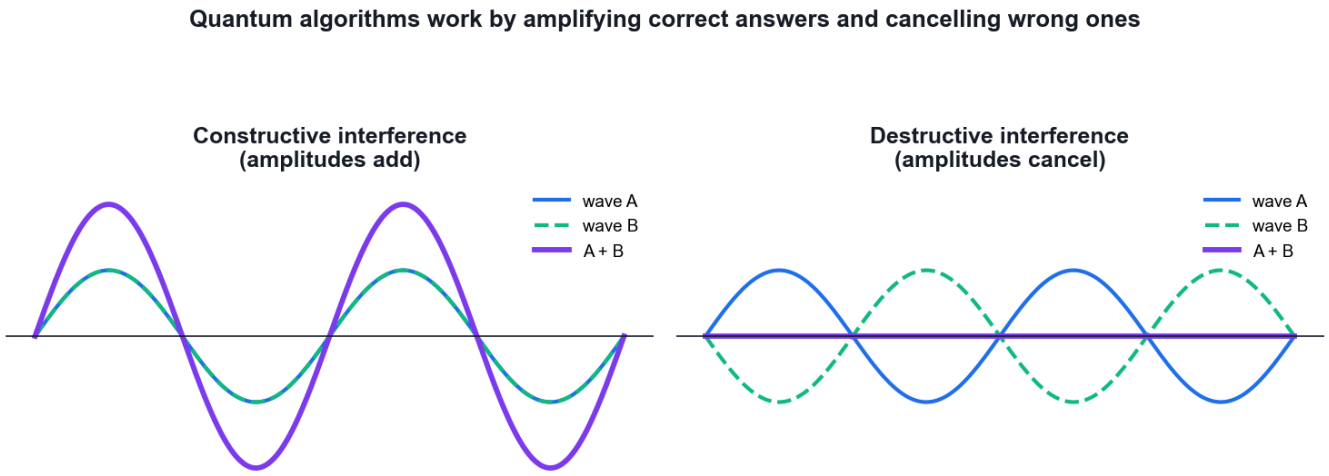
You only ever read out n classical bits — algorithms must be clever.

So

Interference, not brute force, is what gives quantum its advantage.

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Interference — The Real Engine



Amplitudes can cancel

Unlike probabilities, amplitudes are complex and can be negative.

Wrong answers cancel

A good algorithm arranges destructive interference on wrong paths.

Right answers add

Constructive interference concentrates probability on the answer.

Every algorithm

Deutsch, Grover and Shor are all interference engines.

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THE MENTAL MODEL

Superposition supplies the paths. Interference picks the answer.

Every quantum algorithm in this course follows that pattern — prepare a superposition, apply an oracle, then interfere so the correct answer survives.

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COURSE ED-TOOL · QUANTUMLAB BEGINNER

QuantumLab Beginner — Your Practice Environment

https://alfredang.github.io/quantumlabs/

Math Prerequisites & QM Postulates
Vectors, complex numbers, Hermitian, unitary, the 5 axioms

Qubits · Bloch Sphere · Superposition
Interactive Bloch disc, probability slider, live Dirac panel

Entanglement · Bell's Inequality
Bell-pair simulator and a CHSH Monte-Carlo experiment

Circuits · Phase Kickback · Algorithms
Drag-and-drop builder, kickback explorer, Grover/DJ/QFT/Shor

1. Open alfredang.github.io/quantumlabs in any browser — no install.
2. Pick the module matching the topic just taught.
3. Use the interactive simulator to build intuition before the maths.
4. Take the 3-question module quiz to check your understanding.

Built by Tertiary Infotech Academy — 18 interactive modules, runs entirely in the browser, no install.

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18 MODULES · alfredang.github.io/quantumlabs

What QuantumLab Covers

1

Reference modules

Math prerequisites, the five QM postulates, density matrices, history of QM.

2

Core concepts

Qubits, Bloch sphere, superposition, entanglement, measurement.

3

Bell's inequality

CHSH Monte-Carlo comparing the quantum $2\sqrt{2}$ bound against local realism.

4

Circuit builder

Drag-and-drop 2^n state-vector simulator with the full gate set.

5

Phase kickback

Four-state explorer showing the eigenvalue phase moving to the control.

6

Algorithms

Coin flip, Grover, Deutsch-Jozsa, teleportation, QFT and Shor with circuit diagrams.

7

Cryptography

BB84 key distribution with an optional Eve eavesdropper toggle.

8

Noise and QEC

Channel-noise propagation and a 3-qubit bit-flip Monte-Carlo.

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PREREQUISITES

The Mathematics of Quantum Mechanics

Linear algebra · Dirac notation · Unitary operators · The five axioms

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WHY THIS SECTION COMES FIRST

Quantum mechanics is linear algebra with a probability rule bolted on.

Everything in this course follows from five axioms and the properties of unitary and Hermitian operators. Get these right and every algorithm becomes readable.

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Vectors and Dirac notation

A quantum state is a COLUMN VECTOR. Dirac notation names it compactly.

KET $|\psi\rangle$ = column vector $|0\rangle = [1, 0]^T$, $|1\rangle = [0, 1]^T$

BRA $\langle\psi|$ = conjugate transpose = $(|\psi\rangle)^\dagger$ = row vector

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = [\alpha, \beta]^T$

$\langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1| = [\alpha^*, \beta^*]$

KEY POINT: The bra is the DAGGER of the ket: transpose AND complex-conjugate.

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Inner product — overlap and probability

$\langle\varphi|\psi\rangle$ = a single COMPLEX NUMBER (a scalar)

$\langle 0|0\rangle = 1$ (normalised)
$\langle 0|1\rangle = 0$ (orthogonal)
$\langle 0|+\rangle = 1/\sqrt{2}$ (partial overlap)

Properties: $\langle\varphi|\psi\rangle = \langle\psi|\varphi\rangle^*$ and $\langle\psi|\psi\rangle = 1$ for a valid state

KEY POINT: The inner product IS the probability amplitude — square its magnitude to get a probability (the Born rule).

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Outer product — operators and projectors

|ψ><φ| = a MATRIX (an operator), not a number

|0><0| = [[1,0],[0,0]] — projector onto |0>
|1><1| = [[0,0],[0,1]] — projector onto |1>

Completeness: |0><0| + |1><1| = I
Any operator: A = Σ_ij $a_{i\,j}$ |i><j|

KEY POINT: Inner product = number. Outer product = matrix. Getting these two the right way round prevents most beginner errors.

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Tensor product — combining systems

$|a\rangle \otimes |b\rangle$ written $|a\rangle|b\rangle$ or $|ab\rangle$

$|0\rangle \otimes |1\rangle = [1,0]^T \otimes [0,1]^T = [0, 1, 0, 0]^T = |01\rangle$

Dimensions MULTIPLY: $2 \otimes 2 = 4$, and n qubits give 2^n

Operators too: $(A \otimes B)(|a\rangle \otimes |b\rangle) = A|a\rangle \otimes B|b\rangle$

KEY POINT: The tensor product is the mathematical source of exponential state space — and therefore of quantum advantage.

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Eigenvalues and eigenvectors

A|v> = λ|v> |v> is unchanged in DIRECTION, only scaled by λ

Find them by solving the characteristic equation det(A - λI) = 0.

Example — Pauli Z = [[1,0],[0,-1]]:
Z|0> = +1|0> eigenvalue +1, eigenvector |0>
Z|1> = -1|1> eigenvalue -1, eigenvector |1>

KEY POINT: Measurement outcomes ARE eigenvalues, and the measurement basis IS the eigenbasis. Eigenvectors of X are |+> and |->.

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THE CENTRAL PROPERTY

Definition

An operator U is UNITARY when:

$$U^\dagger U = U U^\dagger = I \qquad \text{equivalently} \quad U^{-1} = U^\dagger$$

where $U^\dagger$ is the conjugate transpose (dagger / adjoint):

$$(U^\dagger)_{i\,j} = (U_{j\,i})^*$$

Example — Hadamard: $H = (1/\sqrt{2})[[1,1],[1,-1]]$

$H^\dagger = H$ (real and symmetric), and $H \cdot H = I$ so H is unitary.

KEY POINT: Checking $U^\dagger U = I$ is the single most useful sanity test when you build a gate.

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THE CENTRAL PROPERTY

Property 1 — Unitary preserves the norm

$$\langle U\psi|U\psi\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|I|\psi\rangle = \langle\psi|\psi\rangle = 1$$

So a normalised state STAYS normalised.

Consequence: probabilities always sum to 1, at every step of the circuit.
A non-unitary gate would create or destroy probability — unphysical.

KEY POINT: This is why you never need to renormalise mid-circuit; only measurement (which is not unitary) requires it.

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THE CENTRAL PROPERTY

Property 2 — Unitary preserves inner products

$$\langle U\varphi | U\psi \rangle = \langle \varphi | U^\dagger U | \psi \rangle = \langle \varphi | \psi \rangle$$

ANGLES between states are preserved. Orthogonal states stay orthogonal.

Geometric picture: a unitary is a ROTATION (possibly with reflection) of the state space — it is rigid, it never stretches or squashes.

KEY POINT: This single property is what makes the no-cloning proof work.

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THE CENTRAL PROPERTY

Property 3 — Unitary is reversible

$U^{-1} = U\dagger$ always EXISTS.

So every quantum gate can be undone: $U\dagger(U|\psi\rangle) = |\psi\rangle$

Consequences:

- quantum circuits have equal input and output wire counts
- no information is ever erased (no Landauer heat cost)
- classical AND/OR are NOT valid gates — they lose a bit
- classical logic must be made reversible first (Toffoli gate)

KEY POINT: Reversibility is a constraint on circuit design, not a convenience — you cannot 'discard' a qubit mid-computation without consequences.

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THE CENTRAL PROPERTY

Property 4 — Eigenvalues lie on the unit circle

If U|v> = λ|v>, then norm preservation forces:

|λ| = 1 so λ = e^(iθ) for some real θ

Unitary operators can only apply PHASES to their eigenvectors — never amplify or damp them.

This fact is the entire basis of QUANTUM PHASE ESTIMATION.

KEY POINT: Phase estimation exists precisely because unitary eigenvalues are pure phases: estimating λ reduces to estimating the single real number θ.

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THE CENTRAL PROPERTY

Unitary vs Hermitian — do not confuse them

UNITARY U†U = I describes EVOLUTION (gates)
eigenvalues on the unit circle, e^(iθ)

HERMITIAN A† = A describes OBSERVABLES (measurements)
eigenvalues REAL

The Pauli gates X, Y, Z are BOTH — which is why they serve as gates and as measurement observables. H is also both.

Link: A Hermitian => U = e^(iA) is unitary.

KEY POINT: A common exam question. Unitary = evolution; Hermitian = observable; the Paulis happen to be both.

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BEYOND PURE STATES

Why state vectors are not enough

A ket $|\psi\rangle$ describes a PURE state — full knowledge of the system.

But real systems are often:

- statistical mixtures (the source emits $|0\rangle$ or $|1\rangle$ at random)
- entangled with an environment we cannot access
- one half of an entangled pair

None of these can be written as a single ket.

KEY POINT: You need the density matrix as soon as you discuss noise, decoherence or subsystems — which is to say, any real device.

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Definition

PURE state: $\rho = |\psi\rangle\langle\psi|$

MIXED state: $\rho = \Sigma_i p_i |\psi_i\rangle\langle\psi_i|$ with $\Sigma p_i = 1$

Properties of any valid density matrix:

$\rho^\dagger = \rho$ (Hermitian)

$Tr(\rho) = 1$ (probabilities sum to 1)

$\rho \geq 0$ (positive semi-definite)

KEY POINT: The density matrix generalises the state vector — every ket has a ρ, but not every ρ has a ket.

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BEYOND PURE STATES

Distinguishing pure from mixed

$\mathrm{Tr}(\rho^2) = 1 \iff$ PURE
$\mathrm{Tr}(\rho^2) < 1 \iff$ MIXED

The maximally mixed single-qubit state:
$\rho = I/2 = [[0.5, 0],[0, 0.5]]$ $\mathrm{Tr}(\rho^2) = 0.5$

This carries NO information — every measurement gives 50/50 in every basis.

KEY POINT: Bob's qubit before he receives the classical bits in teleportation is exactly I/2 — which is why no faster-than-light signalling occurs.

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BEYOND PURE STATES

Superposition vs mixture

```
SUPERPOSITION  |+> = (|0>+|1>)/√2
   ρ = [[0.5, 0.5],[0.5, 0.5]]      OFF-DIAGONALS PRESENT
   Measure in X basis: always |+>   — deterministic!

MIXTURE  '50% |0>, 50% |1>'
   ρ = [[0.5, 0],[0, 0.5]]          OFF-DIAGONALS ZERO
   Measure in X basis: 50/50        — random

In the Z basis BOTH give 50/50 — they look identical.
```

KEY POINT: The off-diagonal terms are the COHERENCES. Decoherence is precisely the process that decays them, turning a superposition into a mixture.

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The Five Postulates — Overview

- Postulate 1 — STATE SPACE: a state is a unit vector in a complex Hilbert space.
- Postulate 2 — TIME EVOLUTION: closed systems evolve by a unitary operator.
- Postulate 3 — OBSERVABLES: measurable quantities are Hermitian operators.
- Postulate 4 — BORN RULE: probability is the squared magnitude of the amplitude.
- Postulate 5 — COLLAPSE: measurement projects the state onto the observed eigenspace.

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Postulate 1 — State Space

The state of an isolated quantum system is a UNIT VECTOR in a complex Hilbert space H.

$|\psi\rangle \in H, \quad \text{with} \quad \langle\psi|\psi\rangle = 1$

A single qubit: $H = C^2$, spanned by $|0\rangle$ and $|1\rangle$

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in C, \quad |\alpha|^2 + |\beta|^2 = 1$

n qubits: $H = C^2 \otimes C^2 \otimes \dots \otimes C^2 = C^{(2^n)}$

WHAT IT MEANS

- The amplitudes α and β are COMPLEX numbers, not probabilities.
- Normalisation $\langle\psi|\psi\rangle = 1$ says the total probability is 1.
- A GLOBAL phase carries no physics: $|\psi\rangle$ and $e^{(i\varphi)}|\psi\rangle$ are the same state.
- A RELATIVE phase between terms is physical — it drives interference.

WHY IT MATTERS: This is why n qubits hold 2^n amplitudes: the state space is a TENSOR PRODUCT, so dimensions multiply rather than add. 300 qubits carry more amplitudes than there are atoms in the observable universe.

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Postulate 2 — Time Evolution (Unitarity)

A closed quantum system evolves by a UNITARY operator.

$$|\psi(t)\rangle = U\,|\psi(0)\rangle, \qquad \text{where} \quad U^\dagger U = U U^\dagger = I$$

Equivalently, from the Schrödinger equation:

$$i\hbar\, d|\psi\rangle/dt = H|\psi\rangle \qquad \Rightarrow \qquad U = e^{(-iHt/\hbar)}$$

with H the Hamiltonian (Hermitian, so U is unitary).

WHAT IT MEANS

- EVERY quantum gate is unitary — this is not a design choice but a law.
- Unitary => REVERSIBLE: $U^{-1} = U^\dagger$ always exists, so no information is destroyed.
- Unitary => NORM-PRESERVING: probabilities still sum to 1 after the operation.
- There is no quantum AND or OR gate — those are irreversible and lose information.

WHY IT MATTERS: Reversibility is why quantum circuits have the same number of output wires as input wires, and why classical logic must be embedded reversibly (Toffoli) before a quantum computer can run it.

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THE FIVE AXIOMS OF QUANTUM MECHANICS

Postulate 3 — Observables

Every measurable physical quantity is represented by a HERMITIAN operator.

A observable <=> A† = A

Spectral decomposition: $A = \Sigma_i \lambda_i |i\rangle\langle i|$

λ_i — real eigenvalues = the possible measurement OUTCOMES
$|i\rangle$ — orthonormal eigenvectors = the measurement BASIS

Example: $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, eigenvalues +1 and -1

WHAT IT MEANS

- Hermitian guarantees REAL eigenvalues — measurement results are real numbers.
- Eigenvectors of a Hermitian operator are orthogonal and form a complete basis.
- Measuring 'in the Z basis' means measuring the observable Z.
- Non-commuting observables (like X and Z) cannot be measured simultaneously.

WHY IT MATTERS: The uncertainty principle follows directly: if [A,B] ≠ 0 the two observables have no common eigenbasis, so no state has definite values for both. That is why BB84's two bases are conjugate and why eavesdropping is detectable.

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Postulate 4 — The Born Rule

The probability of an outcome is the SQUARED MAGNITUDE of its amplitude.

$$P(\text{outcome } i) = |\langle i|\psi\rangle|^2 = \langle\psi|P_i|\psi\rangle \quad \text{with } P_i = |i\rangle\langle i|$$

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2, \quad |\alpha|^2 + |\beta|^2 = 1$$

Expectation value of an observable A:

$$\langle A\rangle = \langle\psi|A|\psi\rangle = \Sigma_i \lambda_i P(i)$$

WHAT IT MEANS

- Amplitudes ADD (and can cancel); probabilities never cancel.
- This squaring is what makes interference possible — the source of quantum advantage.
- The rule is fundamentally PROBABILISTIC: identical states can give different results.
- To estimate a probability you must repeat the circuit many times ('shots').

WHY IT MATTERS: The Born rule is why quantum algorithms must ENGINEER interference: you cannot read amplitudes directly, so you must arrange for wrong answers to cancel and the right answer to be amplified before measuring.

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Postulate 5 — Measurement Collapse

Measurement PROJECTS the state onto the observed eigenspace and renormalises.

If outcome i is observed, the post-measurement state is:

$$|\psi'\rangle = P_i|\psi\rangle / \sqrt{(\langle\psi|P_i|\psi\rangle)}$$

Example: $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, measure Z and get 0
$|\psi'\rangle = |0\rangle$ — the superposition is GONE

Measuring again immediately returns 0 with probability 1.

WHAT IT MEANS

- Measurement is IRREVERSIBLE — the only non-unitary operation in quantum mechanics.
- Information about the other branches is destroyed permanently.
- You cannot 'peek' at a quantum state without disturbing it.
- This is why you measure ONCE, at the very end of a quantum circuit.

WHY IT MATTERS: Collapse is the security mechanism of QKD (Eve's measurement disturbs the state) AND the central obstacle in error correction (you must diagnose an error WITHOUT measuring the data — hence syndrome measurement on ancillas).

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KEEP THIS FOR THE OPEN-BOOK ASSESSMENT

The Five Postulates — Reference Card

#	Postulate	Mathematics	Governs
1	State space	$\|\psi\rangle \in H,\ \langle\psi\|\psi\rangle = 1$	What a state IS
2	Time evolution	$\|\psi'\rangle = U\|\psi\rangle,\ U^\dagger U = I$	How states CHANGE
3	Observables	$A^\dagger = A,$ real eigenvalues	What can be MEASURED
4	Born rule	$P(i) = \|\langle i\|\psi\rangle\|^2$	How LIKELY each outcome is
5	Collapse	$\|\psi'\rangle = P_i\|\psi\rangle / \sqrt{\langle\psi\|P_i\|\psi\rangle}$	What happens AFTER measuring

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01

TOPIC 01

Quantum Physics Fundamentals

Postulates of Quantum Mechanics · Mathematics for Quantum Mechanics

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MAPS TO LO1

Key Concepts — Quantum Physics Fundamentals

1

The five postulates

State space, unitary time evolution, observables, the Born rule and wave-function collapse are the axioms every quantum algorithm obeys.

2

Hilbert space

Quantum states are unit vectors in a complex inner-product space; a single qubit lives in a 2-dimensional Hilbert space.

3

Unitary operators

U†U = I preserves the norm — this is why every quantum gate is reversible and probability is conserved.

4

Hermitian operators

A† = A guarantees real eigenvalues, so Hermitian operators represent physically measurable observables.

5

Dirac notation

Bra-ket notation compactly expresses states $|\psi\rangle$, dual vectors $\langle\psi|$, inner products $\langle\varphi|\psi\rangle$ and outer products $|\psi\rangle\langle\varphi|$.

6

Tensor products

The state space of a multi-qubit system is the tensor product of the single-qubit spaces — dimension grows as 2^n.

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THE THREE OBJECTS YOU NEED

The Mathematical Toolkit

State vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A qubit is a unit vector in a 2-dimensional Hilbert space, with $|\alpha|^2 + |\beta|^2 = 1$.

Unitary operator

$$U^\dagger U = I$$

Every quantum gate is unitary — reversible, and probability-preserving.

Hermitian operator

$$A^\dagger = A$$

Observables are Hermitian, which guarantees real measured eigenvalues.

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BRA-KET NOTATION

Dirac Notation at a Glance

Ket / Bra

$$|\psi\rangle \quad \text{and} \quad \langle\psi|$$

A ket is a column vector; a bra is its conjugate transpose (row vector).

Inner product

$$\langle\varphi|\psi\rangle$$

A single complex number — the overlap, or probability amplitude, between two states.

Born rule

$$P(x) = |\langle x|\psi\rangle|^2$$

The probability of measuring outcome x is the squared magnitude of the amplitude.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Physics Fundamentals

Activity 1

- Linear Algebra for Quantum Mechanics in Python

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TOPIC 01 · HANDS-ON ACTIVITY

ACTIVITY 1

Linear Algebra for Quantum Mechanics in Python

Work with the mathematical objects every quantum algorithm is built from: state vectors, inner and outer products, unitary and Hermitian operators, eigenvalues and tensor products — all in NumPy.

MAPS TO LO1 — Design mathematical models required for quantum computing applications

YOU'LL PRODUCE

A Python notebook that verifies unitarity and Hermiticity, computes eigenvalues/eigenvectors, and builds two-qubit states with the tensor product.

Tools: Python 3.13, NumPy, Google Colab / VS Code

Full step-by-step guide: Learner Guide — Activity 1: Linear Algebra for Quantum Mechanics in Python

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ACTIVITY 1 · VERIFY & DEBRIEF

Linear Algebra for Quantum Mechanics in Python

TEST IT — HOW YOU KNOW IT WORKED

H-dagger H equals the identity, X and Z return real eigenvalues +1 and -1, the Born-rule probability of measuring 0 in |+> is 0.5, and the tensor product produces a 4-component two-qubit state.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Physics Fundamentals

- The five postulates
- Hilbert space
- Unitary operators
- Hermitian operators
- Dirac notation
- Tensor products

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02

TOPIC 02

Introduction to Quantum Computing Systems

Overview of quantum computers · Commercial systems · Superposition and Entanglement

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MAPS TO LO2, LO4

Key Concepts — Introduction to Quantum Computing Systems

1

Qubit vs classical bit

A bit is 0 or 1; a qubit is $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, allowing superposition of both values.

2

Bloch sphere

Every pure single-qubit state maps to a point on the unit sphere — the standard geometric picture of qubit state and rotation.

3

Physical qubits

Superconducting (Josephson junction), trapped-ion and topological qubits trade off speed, coherence time and connectivity.

4

Commercial platforms

IBM Quantum, Google, IonQ, Rigetti and D-Wave offer cloud access to real quantum processors today.

5

Superposition

A quantum register of n qubits holds a superposition of all 2^n basis states simultaneously — the source of quantum parallelism.

6

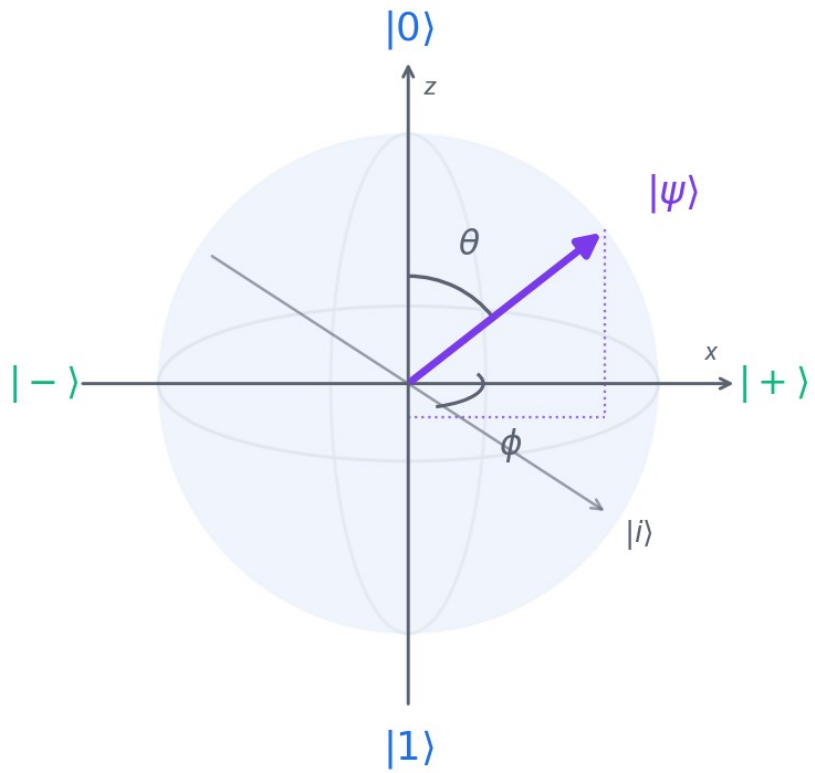
Entanglement

Entangled states cannot be written as a product of single-qubit states; measuring one qubit instantly correlates with the other.

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GEOMETRY OF A SINGLE QUBIT

Visualising a Qubit — The Bloch Sphere



$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$$

North / south poles

|0⟩ and |1⟩ — the classical computational basis states.

The equator

Equal superpositions such as |+⟩ and |−⟩ — maximally uncertain in Z.

θ sets probability

$\cos^2(\theta/2)$ is the chance of measuring 0.

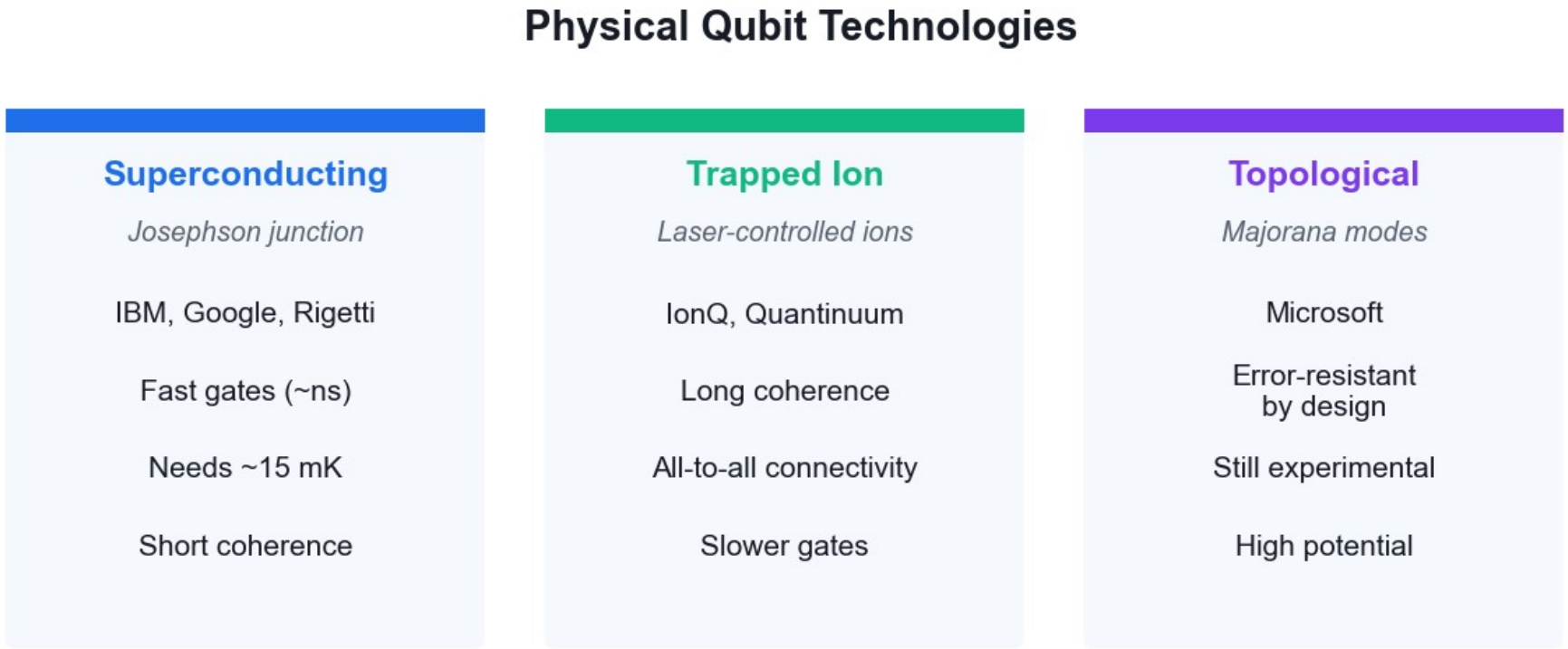
φ sets phase

Invisible to a Z measurement, but it drives interference.

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HOW QUBITS ARE ACTUALLY BUILT

Physical Qubit Technologies

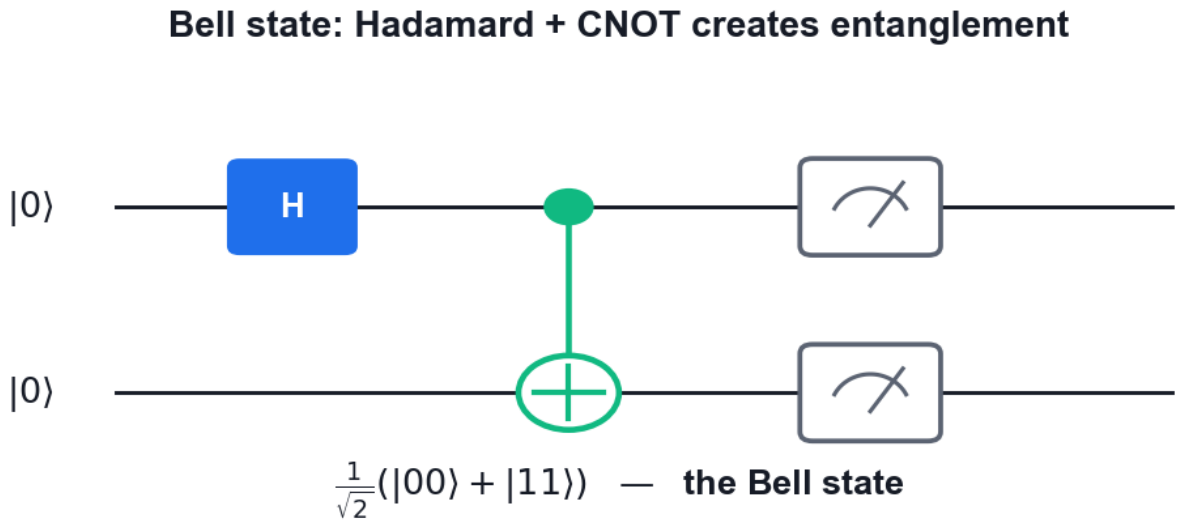


No technology has won yet — each trades gate speed against coherence time and connectivity.

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THE MOST IMPORTANT TWO-QUBIT STATE

Entanglement — The Bell State



Two gates
A Hadamard followed by a CNOT is all it takes.

Perfect correlation
Measure 0 on one qubit and the other is certainly 0.

Not separable
The state cannot be written as a product of two single-qubit states.

The resource
Teleportation, superdense coding and QKD all consume entanglement.

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WHAT YOU'LL DO

Hands-On Activities — Introduction to Quantum Computing Systems

Activity 2

- Visualising a Qubit on the Bloch Sphere

Activity 3

- Superposition and Entanglement on IBM Quantum

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TOPIC 02 · HANDS-ON ACTIVITY

ACTIVITY 2

Visualising a Qubit on the Bloch Sphere

Build single-qubit states in Qiskit and plot them on the Bloch sphere, so you can see how gates rotate a qubit and what superposition and phase actually look like geometrically.

MAPS TO LO2 — Decide on appropriate physical methodologies specific to quantum computing problems

YOU'LL PRODUCE

A set of Bloch-sphere plots showing |0>, |1>, |+>, |-> and an arbitrary Ry-rotated state.

Tools: Qiskit 2.x, matplotlib, Google Colab

Full step-by-step guide: Learner Guide — Activity 2: Visualising a Qubit on the Bloch Sphere

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ACTIVITY 2 · VERIFY & DEBRIEF

Visualising a Qubit on the Bloch Sphere

TEST IT — HOW YOU KNOW IT WORKED

|0> sits at the north pole, X moves it to the south pole, H places it on the equator at +x, and the S gate rotates it around the equator while leaving the Z-basis probabilities at 50/50.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 02 · HANDS-ON ACTIVITY

ACTIVITY 3

Superposition and Entanglement on IBM Quantum

Create your IBM Quantum account, build a superposition circuit and a two-qubit Bell state, and run them on a real quantum processor to see genuine quantum randomness and correlation in the results.

MAPS TO LO2, LO4 — Evaluate quantum computing applicability in the business context

YOU'LL PRODUCE

A Bell state circuit executed on real IBM Quantum hardware, with a measured histogram showing the |00> and |11> correlation.

Tools: IBM Quantum Platform, Qiskit Runtime, Python 3.13

Full step-by-step guide: Learner Guide — Activity 3: Superposition and Entanglement on IBM Quantum

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ACTIVITY 3 · VERIFY & DEBRIEF

Superposition and Entanglement on IBM Quantum

TEST IT — HOW YOU KNOW IT WORKED

The superposition circuit gives roughly 50% |0> and 50% |1>; the Bell circuit gives roughly 50% |00> and 50% |11> with only a small percentage of |01> and |10> counts caused by hardware noise.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Introduction to Quantum Computing Systems

- Qubit vs classical bit
- Bloch sphere
- Physical qubits
- Commercial platforms
- Superposition
- Entanglement

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03

TOPIC 03

Quantum Gates and Circuits

Single Qubit Gates · Multi Qubit Gates · Bell and GHZ States · Phase Kickback

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MAPS TO LO3, LO5

Key Concepts — Quantum Gates and Circuits

1

Pauli gates

X (bit flip), Y and Z (phase flip) are self-inverse, Hermitian and unitary — the building blocks of single-qubit logic.

2

Hadamard gate

H maps $|0\rangle \rightarrow |+\rangle$ and $|1\rangle \rightarrow |-\rangle$, creating the equal superposition that starts nearly every quantum algorithm.

3

Rotation and phase gates

Rx, Ry, Rz rotate the Bloch vector; S and T are special cases of the phase gate $P(\theta)$.

4

Multi-qubit gates

CNOT, CZ, controlled-phase, SWAP and Toffoli (CCNOT) create the entanglement that makes quantum computing powerful.

5

Bell and GHZ states

H followed by CNOT produces a maximally entangled Bell pair; extending to n qubits gives the GHZ state.

6

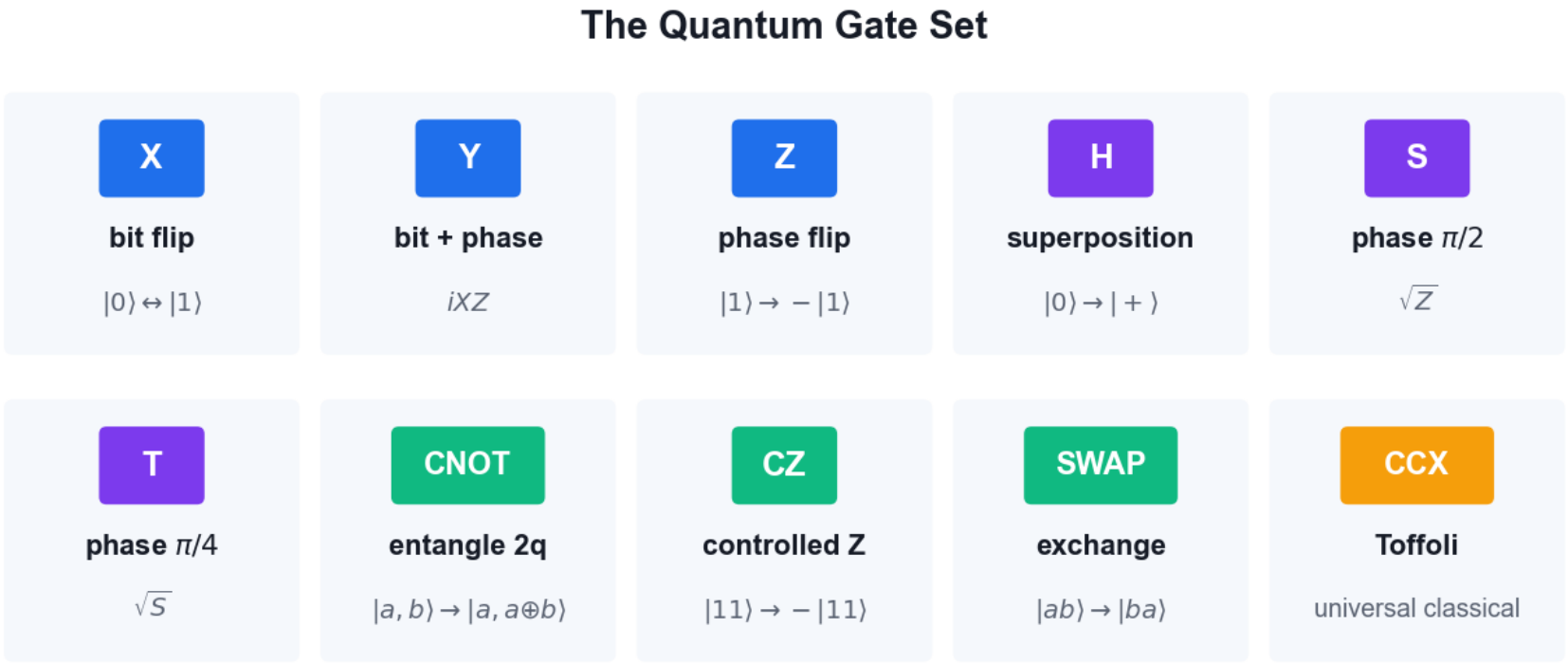
Phase kickback

A controlled operation on an eigenstate transfers the eigenvalue phase back to the control qubit — the engine of Deutsch, BV and QPE.

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YOUR GATE VOCABULARY

The Quantum Gate Set



Single-qubit gates rotate the Bloch vector; two-qubit gates create the entanglement that makes quantum computing powerful.

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GATE REFERENCE

Single vs Multi-Qubit Gates

Gate	Qubits	Effect	Used for
X / Y / Z	1	Rotate π about an axis	Bit flip, phase flip
H	1	Basis change	Create superposition
S / T	1	Phase π/2 and π/4	Fine phase control
Rx / Ry / Rz	1	Arbitrary rotation	Variational ansätze
CNOT	2	Flip target if control is 1	Entanglement, oracles
CZ / CP	2	Conditional phase	QFT, phase kickback
SWAP	2	Exchange two qubits	Routing on hardware
Toffoli	3	Flip if both controls are 1	Reversible classical logic

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THE KEY MECHANISM

Phase kickback is the trick behind almost every algorithm.

Apply a controlled-U to an eigenstate of U and the eigenvalue phase appears on the CONTROL qubit — that is how Deutsch, Bernstein-Vazirani, phase estimation and Shor all extract their answers.

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DERIVATION

Proof — Phase Kickback

CLAIM

A controlled-U acting on an EIGENSTATE of U leaves the target unchanged and transfers the eigenvalue phase onto the CONTROL qubit.

GIVEN

U is a unitary operator with eigenvector |u> and eigenvalue e^(iθ): U|u> = e^(iθ)|u>.

The control qubit starts in |0>; the target starts in the eigenstate |u>.

CU is the controlled-U gate: it applies U to the target only when the control is |1>.

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Step 1 — Put the control into superposition

Start: $|0\rangle \otimes |u\rangle$

Apply H to the control qubit:

$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$

State: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |u\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle|u\rangle + |1\rangle|u\rangle)$

WHY THIS STEP: The control is now in an equal superposition; the target still holds the eigenstate.

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Step 2 — Apply the controlled-U

CU acts term by term (linearity):

CU |0>|u> = |0>|u> (control is 0 -> do nothing)

CU |1>|u> = |1> U|u> (control is 1 -> apply U)

Now use the eigenvalue relation U|u> = e^(iθ)|u>:

CU |1>|u> = |1> e^(iθ)|u> = e^(iθ) |1>|u>

WHY THIS STEP: The scalar e^(iθ) can be moved freely — it multiplies the WHOLE term.

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Step 3 — Collect the terms

State: $(1/\sqrt{2})(|0\rangle|u\rangle + e^{(i\theta)} |1\rangle|u\rangle)$
Factor the target $|u\rangle$ out of both terms:

$$= (1/\sqrt{2})(|0\rangle + e^{(i\theta)}|1\rangle) (x) |u\rangle$$

WHY THIS STEP: This factorisation is the heart of the proof — the state is SEPARABLE again.

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Step 4 — Read off the result

Target register: $|u\rangle$ — UNCHANGED
Control register: $(|0\rangle + e^{i\theta}|1\rangle)/\sqrt{2}$

The phase $e^{i\theta}$, which belonged to the TARGET's eigenvalue, now sits as a RELATIVE PHASE on the CONTROL qubit.

WHY THIS STEP: The information moved from the target to the control without the target changing.

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Step 5 — The oracle special case

Take U = X (the NOT gate) and |u> = |-> = (|0> - |1>)/√2.
Check it is an eigenstate: X|-> = (X|0> - X|1>)/√2
= (|1> - |0>)/√2
= -(|0> - |1>)/√2 = (-1)|->
So |-> is an eigenvector of X with eigenvalue -1, i.e. e^(iπ).

For an oracle U_f: |x>|y> -> |x>|y ⊕ f(x)>, setting the target to |->:
U_f |x>|-> = (-1)^f(x) |x>|->

WHY THIS STEP: This is THE identity that powers Deutsch, Deutsch-Jozsa, Bernstein-Vazirani, Grover's oracle and quantum phase estimation.

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QED

Phase Kickback — Result

RESULT

Controlled-U on an eigenstate |u> gives (|0> + e^(iθ)|1>)/√2 (x) |u> — the eigenvalue phase is 'kicked back' onto the control. With U = X and target |->, the oracle marks inputs with the sign (-1)^f(x).

WHY IT MATTERS

Measurement cannot see a GLOBAL phase, but a RELATIVE phase on the control is converted into a measurable amplitude by a final Hadamard. Phase kickback is how a quantum computer extracts a global property of f without evaluating f at every input.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Gates and Circuits

Activity 4

- Building Circuits in IBM Quantum Composer

Activity 5

- Quantum Gates and Bell/GHZ States with PennyLane

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TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 4

Building Circuits in IBM Quantum Composer

Use the drag-and-drop IBM Quantum Composer to build single- and multi-qubit circuits, watch the state vector and probability panels update live, and export the circuit as OpenQASM.

MAPS TO LO5 — Spearhead the application of quantum gates to quantum computing domains

YOU'LL PRODUCE

A set of Composer circuits (X, H, CNOT, Bell state) with their statevector output, exported as OpenQASM 3 code.

Tools: IBM Quantum Composer, OpenQASM 3

Full step-by-step guide: Learner Guide — Activity 4: Building Circuits in IBM Quantum Composer

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ACTIVITY 4 · VERIFY & DEBRIEF

Building Circuits in IBM Quantum Composer

TEST IT — HOW YOU KNOW IT WORKED

The Composer probability panel shows a 50/50 split between |00> and |11> for the Bell circuit, and the exported OpenQASM contains an h instruction followed by a cx instruction.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 5

Quantum Gates and Bell/GHZ States with PennyLane

Use PennyLane's quantum-function and QNode model to apply the full single- and multi-qubit gate set, then construct Bell and GHZ entangled states and verify their measurement correlations.

MAPS TO LO3, LO5 — Direct two-level quantum models and apply quantum gates

YOU'LL PRODUCE

A PennyLane script that draws each circuit, returns the state vector, and confirms the Bell and GHZ correlations.

Tools: PennyLane, Python 3.13, matplotlib

Full step-by-step guide: Learner Guide — Activity 5: Quantum Gates and Bell/GHZ States with PennyLane

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ACTIVITY 5 · VERIFY & DEBRIEF

Quantum Gates and Bell/GHZ States with PennyLane

TEST IT — HOW YOU KNOW IT WORKED

The Bell state returns amplitudes of about 0.707 on |00> and |11> and zero elsewhere; the GHZ state returns amplitudes only on |000> and |111>; the phase-kickback circuit shows the eigenvalue phase transferred to the control qubit.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Gates and Circuits

- Pauli gates
- Hadamard gate
- Rotation and phase gates
- Multi-qubit gates
- Bell and GHZ states
- Phase kickback

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04

TOPIC 04

Quantum Protocols

No-Cloning Theorem · Teleportation · Superdense Coding · BB84 QKD · CHSH Game

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MAPS TO LO3, LO7

Key Concepts — Quantum Protocols

1

No-cloning theorem

No unitary can copy an arbitrary unknown quantum state — proved by linearity, and the foundation of quantum key security.

2

Quantum teleportation

Transfers an unknown qubit state using one shared Bell pair plus two classical bits — no faster-than-light signalling.

3

Superdense coding

The dual of teleportation: send two classical bits by transmitting a single qubit of a pre-shared entangled pair.

4

BB84 QKD

Alice and Bob build a shared secret key using random bases; eavesdropping disturbs the states and raises the error rate.

5

Eavesdropping detection

An intercept-resend attack introduces roughly a 25% error rate in the sifted key, revealing Eve's presence.

6

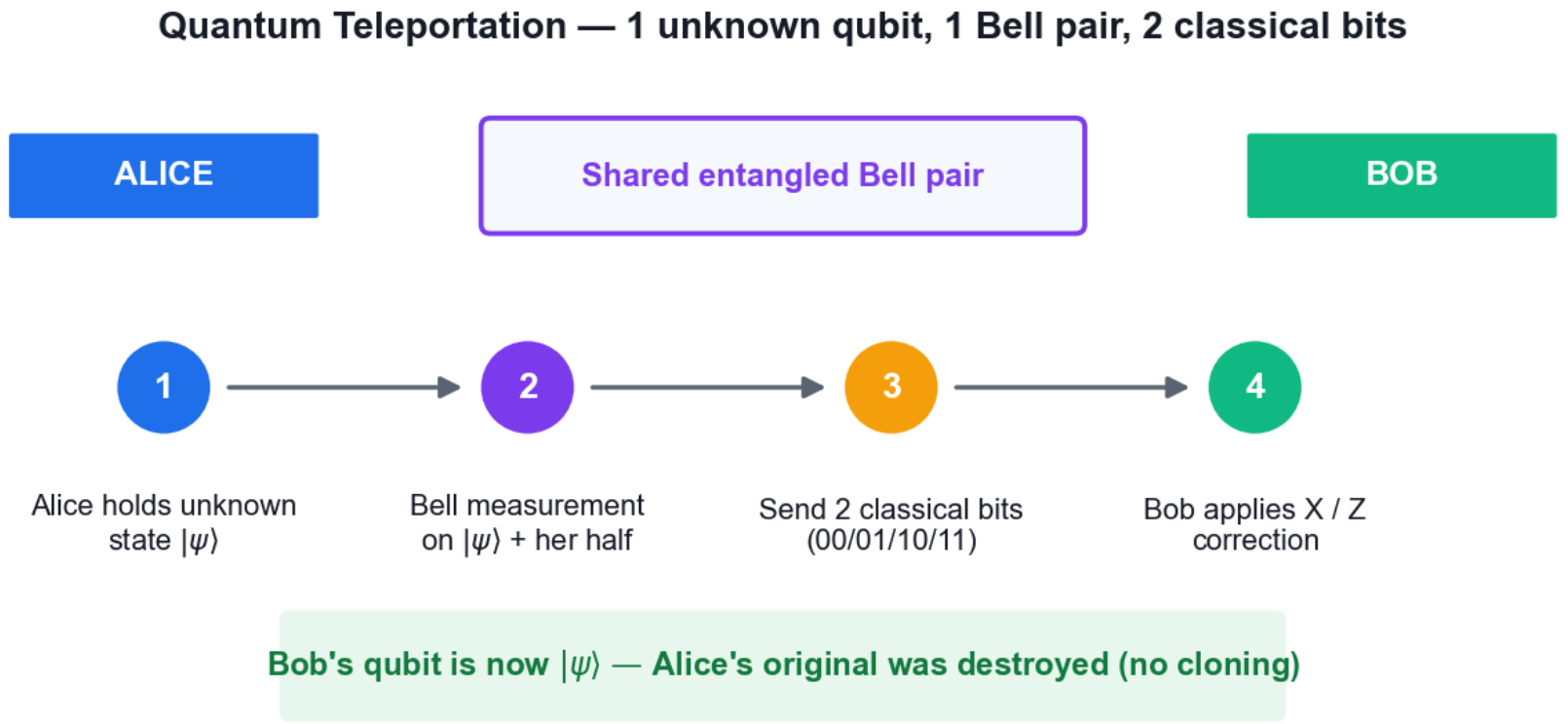
CHSH game

A quantum strategy using entanglement wins ~85% of the time versus the classical maximum of 75%, disproving local hidden variables.

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MOVING A STATE WITHOUT MOVING A PARTICLE

Quantum Teleportation



Two classical bits must still travel — so teleportation never beats the speed of light.

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TWO SIDES OF ONE COIN

Teleportation vs Superdense Coding

Qubit → classical channel

- Quantum Teleportation
  - Sends: 1 unknown QUBIT state
  - Costs: 1 Bell pair + 2 classical bits
  - Original is destroyed (no cloning)
  - Use: moving quantum data

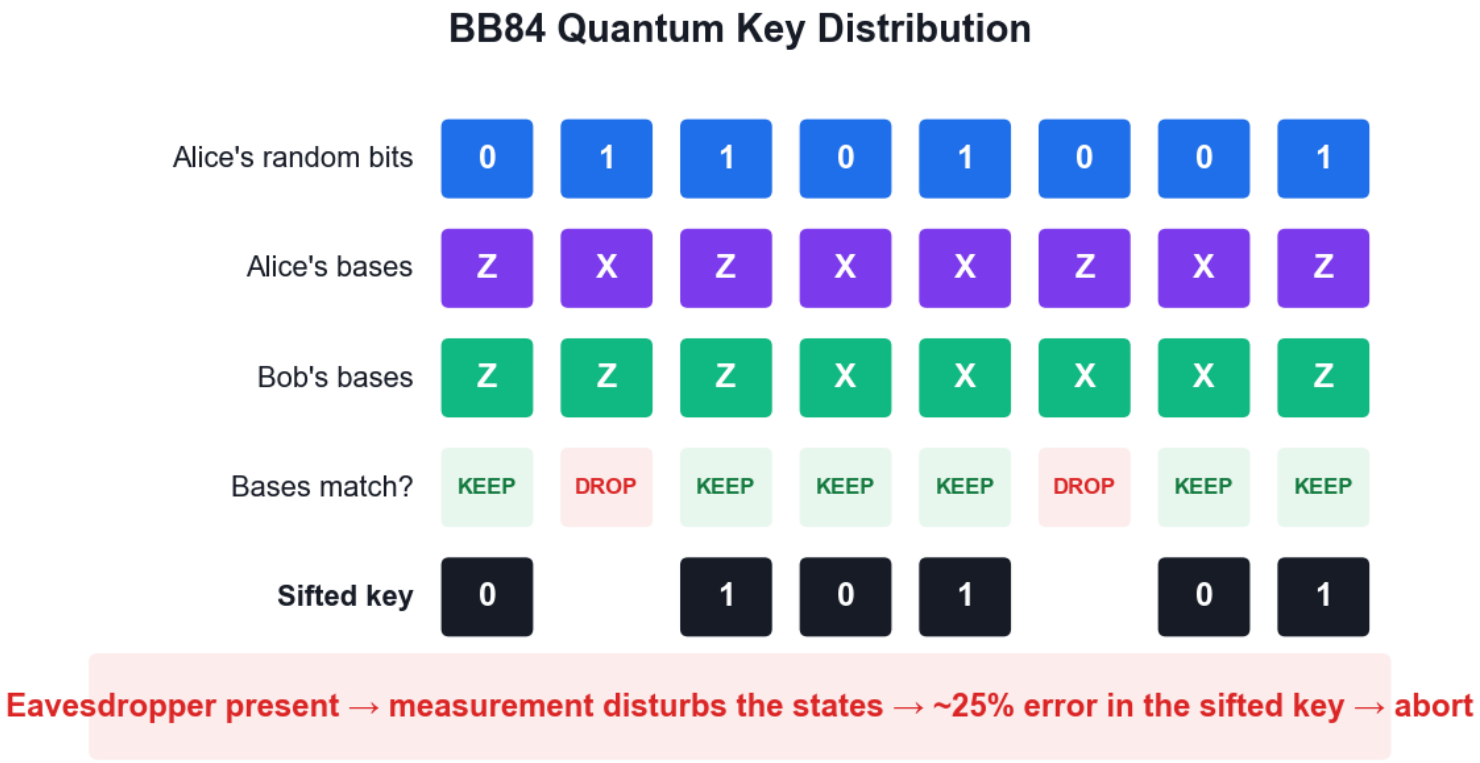
Classical → qubit channel

- Superdense Coding
  - Sends: 2 CLASSICAL bits
  - Costs: 1 Bell pair + 1 transmitted qubit
  - Encode with I, X, Z or ZX
  - Use: doubling channel capacity

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SECURITY FROM PHYSICS, NOT MATHS

BB84 Quantum Key Distribution



Security rests on the no-cloning theorem — an eavesdropper cannot copy the qubits without disturbing them.

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DERIVATION

Proof — Quantum Teleportation

CLAIM

An unknown qubit state can be transferred using one shared Bell pair and two classical bits — without ever measuring or copying the state itself.

GIVEN

Alice holds an unknown qubit $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2 + |b|^2 = 1$.

Alice and Bob share the Bell pair $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

Qubit ordering: q0 = Alice's unknown state, q1 = Alice's half, q2 = Bob's half.

Neither party knows a or b — and neither learns them at any point.

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Step 1 — Write the full three-qubit state

$$|\Psi_0\rangle = |\psi\rangle_0 \otimes |\Phi^+\rangle_{12}$$
$$= (a|0\rangle + b|1\rangle) \otimes (|00\rangle + |11\rangle)/\sqrt{2}$$

Expand:

$$= (1/\sqrt{2})(a|000\rangle + a|011\rangle + b|100\rangle + b|111\rangle)$$

WHY THIS STEP: Three qubits, so the state space has $2^3 = 8$ dimensions; four terms are populated.

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Step 2 — Alice applies CNOT (q0 -> q1)

CNOT flips q1 when q0 = 1. Term by term:

a|000> -> a|000> (q0=0, unchanged)
a|011> -> a|011> (q0=0, unchanged)
b|100> -> b|110> (q0=1, flip q1)
b|111> -> b|101> (q0=1, flip q1)

$|\Psi_1\rangle = (1/\sqrt{2})(a|000\rangle + a|011\rangle + b|110\rangle + b|101\rangle)$

WHY THIS STEP: This begins Alice's Bell-basis measurement — CNOT then H rotates the Bell basis into the computational basis.

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Step 3 — Alice applies H to q0

H|0> = (|0>+|1>)/√2 , H|1> = (|0>-|1>)/√2. Substituting:

$$|\Psi_2\rangle = (1/2)[\ a(|0\rangle+|1\rangle)|00\rangle + a(|0\rangle+|1\rangle)|11\rangle + b(|0\rangle-|1\rangle)|10\rangle + b(|0\rangle-|1\rangle)|01\rangle\ ]$$

WHY THIS STEP: Now expand and regroup by Alice's two qubits (q0 q1) — the next step.

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Step 4 — Regroup by Alice's measurement outcome

Collecting terms with the same q0q1 prefix:

$$|\Psi_2\rangle = \frac{1}{2}\big[\ |00\rangle(a|0\rangle + b|1\rangle) + |01\rangle(a|1\rangle + b|0\rangle) + |10\rangle(a|0\rangle - b|1\rangle) + |11\rangle(a|1\rangle - b|0\rangle)\ \big]$$

WHY THIS STEP: THIS IS THE KEY LINE. Bob's qubit is already |ψ> up to a known correction, in every branch.

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Step 5 — Alice measures; Bob's state collapses

Each outcome has probability $|1/2|^2 = 1/4$ — independent of a and b.

00 -> Bob has $a|0> + b|1>$ = $|\psi>$
01 -> Bob has $a|1> + b|0>$ = $X|\psi>$
10 -> Bob has $a|0> - b|1>$ = $Z|\psi>$
11 -> Bob has $a|1> - b|0>$ = $XZ|\psi>$ (up to phase)

WHY THIS STEP: Because every outcome is equally likely REGARDLESS of a and b, Alice's measurement reveals nothing about the state — this is why no information leaks.

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Step 6 — Bob applies the correction

Alice sends the two classical bits ($b_0 b_1$). Bob applies:

```
00  ->  I    (do nothing)        ->  |ψ>
01  ->  X                        ->  X·X|ψ>  = |ψ>
10  ->  Z                        ->  Z·Z|ψ>  = |ψ>
11  ->  ZX                        ->          |ψ>
```

using $X^2 = Z^2 = I$.

WHY THIS STEP: In every branch Bob ends holding exactly a|0> + b|1>.

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Step 7 — Consistency checks

NO CLONING

Alice's q0 was measured, so her copy is destroyed.
Only ONE |ψ> exists at the end — no cloning occurred.

NO FASTER-THAN-LIGHT SIGNALLING

Before the classical bits arrive, Bob's qubit is the maximally mixed state ρ = I/2 — it carries no information at all.
The two classical bits travel at or below light speed.

NO STATE KNOWLEDGE

Neither party ever learns the amplitudes a or b.

WHY THIS STEP: Teleportation respects both quantum and relativistic limits.

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QED

Quantum Teleportation — Result

RESULT

Bob's qubit is exactly |ψ> = a|0> + b|1> after applying the correction indicated by Alice's two classical bits. The original was destroyed by the measurement.

WHY IT MATTERS

Teleportation is the foundation of quantum networking and the quantum internet, and is used inside fault-tolerant architectures to move logical qubits between code blocks. It shows entanglement + classical communication can substitute for sending a qubit.

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DERIVATION

Proof — The No-Cloning Theorem

CLAIM

No unitary operation can copy an arbitrary unknown quantum state.

GIVEN

Suppose a cloning machine U exists with $U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$ for EVERY state $|\psi\rangle$.

$|0\rangle$ is a blank slate; $|\psi\rangle$ is the state to copy.

We derive a contradiction — this is a proof by contradiction.

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Step 1 — Assume two arbitrary states

Let $|\psi\rangle$ and $|\varphi\rangle$ be any two states. By assumption:

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$$
$$U(|\varphi\rangle|0\rangle) = |\varphi\rangle|\varphi\rangle$$

WHY THIS STEP: No restriction yet — this is just the definition of a universal cloner.

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Step 2 — Unitary preserves inner products

For any unitary U: $\langle U\alpha|U\beta\rangle = \langle\alpha|U^\dagger U|\beta\rangle = \langle\alpha|I|\beta\rangle = \langle\alpha|\beta\rangle$

Apply this to our two inputs:

LHS = $\langle\psi|\varphi\rangle \cdot \langle 0|0\rangle = \langle\psi|\varphi\rangle$

RHS = $\langle\psi|\varphi\rangle \cdot \langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$

WHY THIS STEP: Unitarity is the only physical assumption used — this is why the theorem is so general.

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Step 3 — Solve the resulting equation

Setting LHS = RHS and writing $z = \langle\psi|\varphi\rangle$:

$$z = z^2$$
$$z^2 - z = 0$$
$$z(z - 1) = 0$$

So either $z = 0$ or $z = 1$.

WHY THIS STEP: Only two possibilities survive.

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Step 4 — Interpret the two solutions

$z = \langle\psi|\varphi\rangle = 1 \Rightarrow |\psi\rangle$ and $|\varphi\rangle$ are the SAME state.
$z = \langle\psi|\varphi\rangle = 0 \Rightarrow |\psi\rangle$ and $|\varphi\rangle$ are ORTHOGONAL.

But we assumed U clones ARBITRARY states — including states that are neither identical nor orthogonal, e.g. $|0\rangle$ and $|+\rangle$, where $\langle 0|+\rangle = 1/\sqrt{2}$.

$1/\sqrt{2}$ is neither 0 nor 1 -> CONTRADICTION.

WHY THIS STEP: The assumption fails for any pair of non-orthogonal, non-identical states.

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Step 5 — Conclusion

No unitary can clone an arbitrary unknown state.

What IS possible:

- copying states known to be from an ORTHOGONAL set (classical copying)
- MOVING a state (teleportation — the original is destroyed)
- approximate cloning with bounded fidelity (max ~5/6 for a universal cloner)

WHY THIS STEP: Note the theorem forbids perfect copying of UNKNOWN states — classical bits, which are always in an orthogonal set, remain copyable.

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QED

The No-Cloning Theorem — Result

RESULT

Perfect cloning of an arbitrary unknown quantum state is impossible — it would violate the linearity/unitarity of quantum mechanics.

WHY IT MATTERS

No-cloning is the security foundation of QKD: an eavesdropper cannot copy the qubits in transit and must instead measure them, which disturbs the state and reveals her presence. It is also why quantum error correction cannot work by simple redundancy.

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DERIVATION

Proof — BB84 Security and Eavesdropper Detection

CLAIM

An intercept-resend eavesdropper introduces a 25% error rate in the sifted key and is therefore detected with near-certainty.

GIVEN

Two conjugate bases: Z = {|0>, |1>} and X = {|+>, |->}.

Encoding: bit 0 -> |0> or |+> ; bit 1 -> |1> or |-> (basis chosen at random).

The bases are UNBIASED: $|\langle 0|+\rangle|^2 = |\langle 0|-\rangle|^2 = |\langle 1|+\rangle|^2 = |\langle 1|-\rangle|^2 = 1/2$.

Alice and Bob each choose their basis independently and uniformly at random.

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Step 1 — Sifting (no eavesdropper)

Alice sends n qubits. Bob measures each in a random basis.

P(bases match) = 1/2, so about n/2 qubits survive sifting.

When the bases MATCH, Bob's measurement is deterministic:
  Alice sends |0> in Z, Bob measures Z -> gets 0 with probability 1
So the sifted key agrees perfectly (up to channel noise).

WHY THIS STEP: The bits where the bases differed are publicly discarded — they carry no key.

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Step 2 — Eve intercepts and resends

Eve does not know Alice's basis, so she guesses:
P(Eve picks the right basis) = 1/2

CASE 1 — Eve guesses RIGHT (prob 1/2):
She measures correctly, resends the same state.
Bob (same basis as Alice) gets the right bit. NO error introduced.

WHY THIS STEP: Half the time Eve is invisible on that qubit.

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Step 3 — Eve guesses the WRONG basis

CASE 2 — Eve guesses WRONG (prob 1/2):

Say Alice sent |0> (Z basis) but Eve measures in X.
|0> = (|+> + |->)/√2
Eve gets |+> or |-> with probability 1/2 each — and RESENDS that.

Bob now measures in Z (matching Alice). Say Eve resent |+>:
|+> = (|0> + |1>)/√2
Bob gets 0 or 1 with probability 1/2 each.
-> Bob's bit is WRONG with probability 1/2.

WHY THIS STEP: Eve's wrong-basis measurement has destroyed the original state. She cannot restore it — that is the no-cloning theorem doing the work.

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Step 4 — Compute the total error rate

P(error) = P(Eve wrong basis) × P(Bob then errs)
= 1/2 × 1/2
= 1/4 = 25%

So on the sifted key, roughly 1 bit in 4 disagrees between Alice and Bob.

WHY THIS STEP: This 25% figure — the Quantum Bit Error Rate (QBER) — is the signature of intercept-resend.

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Step 5 — Detection probability

Alice and Bob publicly compare a random sample of m sifted bits.

P(a given compared bit fails to reveal Eve) = 1 - 1/4 = 3/4
P(Eve escapes detection over m bits) = (3/4)^m

m = 10 -> 5.6% escape chance
m = 50 -> 0.000006%
m = 100 -> ~3 × 10^{-13}

WHY THIS STEP: Detection probability rises exponentially with the size of the test sample.

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Step 6 — Decision rule and post-processing

Alice and Bob measure the QBER on the test sample:

QBER below threshold (~11%) -> proceed:
- error correction (reconcile the remaining differences)
- privacy amplification (hash down to a shorter, secure key)

QBER above threshold -> ABORT and discard the key

The discarded test bits are never reused as key material.

WHY THIS STEP: Real systems tolerate a few percent QBER from hardware noise; the threshold separates tolerable noise from an active attack.

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QED

BB84 Security and Eavesdropper Detection — Result

RESULT

Intercept-resend produces a 25% QBER. Comparing 100 random sifted bits detects Eve with probability $1 - (3/4)^{100} \approx 1 - 3\times10^{-13}$.

WHY IT MATTERS

BB84's security rests on PHYSICS (no-cloning + measurement disturbance), not on a computational hardness assumption. A quantum computer does not weaken it — unlike RSA and ECC, which Shor's algorithm breaks outright.

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DERIVATION

Proof — The CHSH Inequality and Bell Violation

CLAIM

Local hidden-variable theories obey $|S| \leq 2$, but quantum mechanics reaches $2\sqrt{2} \approx 2.828$.

GIVEN

Alice measures A_0 or A_1; Bob measures B_0 or B_1. Each outcome is ±1.

The CHSH quantity: $S = \langle A_0B_0 \rangle + \langle A_0B_1 \rangle + \langle A_1B_0 \rangle - \langle A_1B_1 \rangle$.

LOCAL REALISM assumes each measurement has a definite pre-existing value.

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Step 1 — The classical bound

Under local realism every $a_0, a_1, b_0, b_1 \in \{+1, -1\}$ exists simultaneously.

$$S = a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1$$
$$= a_0(b_0 + b_1) + a_1(b_0 - b_1)$$

Since $b_0, b_1 = \pm 1$, one of $(b_0 + b_1)$, $(b_0 - b_1)$ is 0 and the other is ± 2.
So $S = \pm 2$ in every case, hence $|S| \leq 2$.

WHY THIS STEP: This is the CHSH inequality. It assumes ONLY locality and realism — no physics beyond that.

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Step 2 — The quantum prediction

Take the singlet $|\Psi-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$.
For measurement directions a and b, quantum mechanics predicts:

$$\langle A(a)B(b)\rangle = -\cos(\theta_{ab})$$

where θ_{ab} is the angle between the two measurement axes.

WHY THIS STEP: This correlation is derived from the Born rule alone.

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Step 3 — Choose the optimal angles

Pick $A_0 = 0°$, $A_1 = 45°$, $B_0 = 22.5°$, $B_1 = 67.5°$. Then:

$\langle A_0B_0 \rangle = -\cos(22.5°) = -0.9239$
$\langle A_0B_1 \rangle = -\cos(67.5°) = -0.3827$
$\langle A_1B_0 \rangle = -\cos(22.5°) = -0.9239$
$\langle A_1B_1 \rangle = -\cos(22.5°) = -0.9239$ (angle 22.5° the other way)

$|S| = |-0.9239 - 0.3827 - 0.9239 + 0.9239| \ldots = 2\sqrt{2} \approx 2.828$

WHY THIS STEP: $2\sqrt{2}$ is the Tsirelson bound — the maximum quantum mechanics allows.

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Step 4 — What the violation means

Classical (local realism): $|S| \leq 2$
Quantum mechanics: $|S| = 2\sqrt{2} \approx 2.828$
Experiment (Aspect 1982, loophole-free 2015): ~2.4-2.8

Nature VIOLATES the classical bound. Therefore no local hidden-variable theory can reproduce quantum mechanics.

2022 Nobel Prize: Aspect, Clauser and Zeilinger.

WHY THIS STEP: Entanglement is not 'hidden classical correlation' — it is a genuinely non-classical resource.

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QED

The CHSH Inequality and Bell Violation — Result

RESULT

Local hidden variables force $|S| \leq 2$; entangled qubits achieve $2\sqrt{2}$. Experiments confirm the quantum value, ruling out local realism.

WHY IT MATTERS

The CHSH violation is the basis of DEVICE-INDEPENDENT cryptography: if your devices violate the inequality, entanglement is genuine and a key can be certified secure even with untrusted hardware. It also underpins the E91 QKD protocol.

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DERIVATION

Proof — Superdense Coding

CLAIM

One qubit of a pre-shared entangled pair carries TWO classical bits.

GIVEN

Alice and Bob share $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ in advance.

Alice wants to send a 2-bit message; she may transmit only ONE qubit.

The four Bell states form an orthonormal basis of the 2-qubit space.

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Step 1 — Alice encodes with a local operation

Alice applies one gate to HER qubit only:

message 00 -> I -> (|00> + |11>)/√2 = |Φ+>
message 01 -> X -> (|10> + |01>)/√2 = |Ψ+>
message 10 -> Z -> (|00> - |11>)/√2 = |Φ->
message 11 -> ZX -> (|10> - |01>)/√2 = |Ψ->

WHY THIS STEP: Alice never touches Bob's qubit — yet the GLOBAL state changes, because the pair is entangled.

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Step 2 — The four results are orthogonal

<Φ+|Φ-> = 0, <Φ+|Ψ+> = 0, <Φ+|Ψ-> = 0, and so on.

All four Bell states are mutually ORTHOGONAL, so they can be distinguished perfectly by a single measurement in the Bell basis.

WHY THIS STEP: Orthogonality is exactly what makes perfect discrimination possible — contrast with BB84, where non-orthogonal states protect the key.

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Step 3 — Bob decodes

Alice sends her single qubit to Bob. He now holds both and undoes the entangling circuit:

CNOT (control = Alice's qubit, target = Bob's), then H on Alice's qubit

|Φ+> -> |00> |Ψ+> -> |01>
|Φ-> -> |10> |Ψ-> -> |11>

Bob measures both qubits and reads the 2-bit message.

WHY THIS STEP: The Bell measurement maps the four Bell states back to the computational basis.

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Step 4 — Accounting and limits

Total qubits transmitted: 1. Classical bits delivered: 2.

This does NOT violate Holevo's bound: the entangled pair had to be distributed in advance, which itself required sending a qubit.

Superdense coding is the DUAL of teleportation:

teleportation: 2 classical bits + 1 ebit -> 1 qubit

superdense coding: 1 qubit + 1 ebit -> 2 classical bits

WHY THIS STEP: Entanglement is a RESOURCE that can be traded against communication.

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QED

Superdense Coding — Result

RESULT

Alice transmits two classical bits by sending a single qubit, using a pre-shared entangled pair and a local Pauli operation.

WHY IT MATTERS

Superdense coding proved entanglement has quantitative communication value, launching quantum information theory as a discipline and motivating entanglement-distribution networks.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Protocols

Activity 6

- Quantum Teleportation

Activity 7

- Superdense Coding Protocol

Activity 8

- BB84 Quantum Key Distribution with Eavesdropping Detection

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TOPIC 04 · HANDS-ON ACTIVITY

ACTIVITY 6

Quantum Teleportation

Implement the three-qubit teleportation protocol: entangle a shared pair, perform a Bell measurement on the unknown state, and apply the conditional corrections that recreate the original state on Bob's qubit.

MAPS TO LO3, LO7 — Synthesize quantum methodologies to address complex problems

YOU'LL PRODUCE

A teleportation circuit run on IBM Quantum hardware that reconstructs an arbitrary input state at the receiving qubit.

Tools: Qiskit 2.x, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 6: Quantum Teleportation

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ACTIVITY 6 · VERIFY & DEBRIEF

Quantum Teleportation

TEST IT — HOW YOU KNOW IT WORKED

The measurement statistics on Bob's qubit q2 match the amplitudes of the state originally prepared on q0, within hardware noise, and the original state no longer exists on q0.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 04 · HANDS-ON ACTIVITY

ACTIVITY 7

Superdense Coding Protocol

Implement the dual of teleportation: transmit two classical bits by sending only a single qubit of a pre-shared entangled pair, encoding the message with the I, X, Z and XZ operations.

MAPS TO LO3, LO7 — Direct two-level quantum models for quantum computing

YOU'LL PRODUCE

A superdense coding circuit that correctly transmits all four two-bit messages 00, 01, 10 and 11.

Tools: Qiskit 2.x, IBM Quantum Runtime

Full step-by-step guide: Learner Guide — Activity 7: Superdense Coding Protocol

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ACTIVITY 7 · VERIFY & DEBRIEF

Superdense Coding Protocol

TEST IT — HOW YOU KNOW IT WORKED

Each of the four encodings decodes to its intended message — identity gives 00, X gives 01, Z gives 10 and ZX gives 11 — demonstrating two classical bits carried by one transmitted qubit.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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ACTIVITY 8

BB84 Quantum Key Distribution with Eavesdropping Detection

Run the BB84 protocol end to end: random bit and basis selection, quantum transmission, basis sifting and key extraction — then repeat it with an eavesdropper present and measure the error rate that exposes the attack.

MAPS TO LO7 — Synthesize quantum methodologies to address complex problems

YOU'LL PRODUCE

A working BB84 implementation producing a shared secret key, plus a comparison run showing the ~25% error rate introduced by an intercept-resend eavesdropper.

Tools: Qiskit 2.x, IBM Quantum Runtime, Python

Full step-by-step guide: Learner Guide — Activity 8: BB84 Quantum Key Distribution with Eavesdropping Detection

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ACTIVITY 8 · VERIFY & DEBRIEF

BB84 Quantum Key Distribution with Eavesdropping Detection

TEST IT — HOW YOU KNOW IT WORKED

Without an eavesdropper the sifted key error rate is near zero (bounded only by hardware noise); with the intercept-resend eavesdropper the error rate rises to roughly 25%, which is the signal to abort and discard the key.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Protocols

- No-cloning theorem
- Quantum teleportation
- Superdense coding
- BB84 QKD
- Eavesdropping detection
- CHSH game

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05

TOPIC 05

Quantum Algorithms

Deutsch · Deutsch-Jozsa · Bernstein-Vazirani · Grover · Quantum Fourier Transform

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MAPS TO LO6, LO7

Key Concepts — Quantum Algorithms

1

The query (oracle) model

Algorithms are compared by how many times they query a black-box function — the standard measure of quantum advantage.

2

Deutsch algorithm

Decides whether $f:\{0,1\}\to\{0,1\}$ is constant or balanced with ONE query; classically two are required.

3

Deutsch-Jozsa

Extends the idea to n bits — one query versus $2^{n-1}+1$ classical queries in the worst case: exponential separation.

4

Bernstein-Vazirani

Recovers an n-bit hidden string s from $f(x) = s\cdot x \pmod 2$ in a single query instead of n.

5

Grover search

Amplitude amplification finds a marked item in an unstructured database in $O(\sqrt{N})$ instead of $O(N)$ — a quadratic speedup.

6

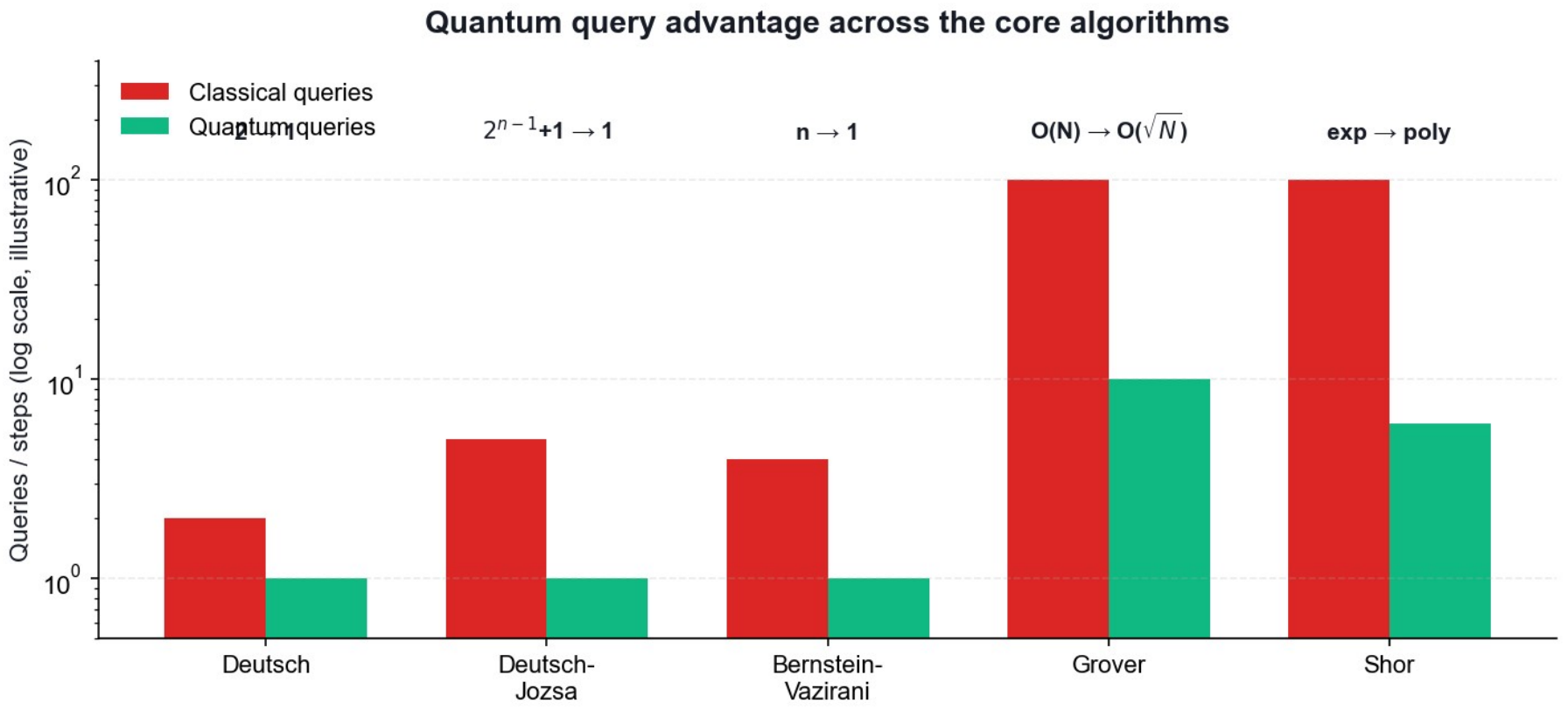
Quantum Fourier Transform

The quantum analogue of the DFT runs in $O(n^2)$ gates and underpins phase estimation and Shor's algorithm.

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WHY THESE ALGORITHMS MATTER

Quantum Query Advantage

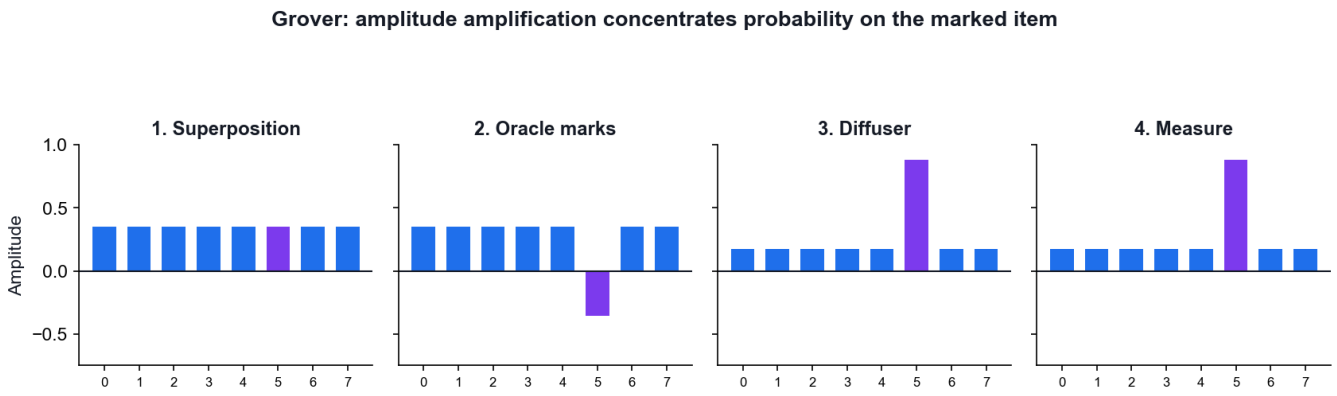


Query counts are illustrative — the point is the SCALING, not the absolute numbers.

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HOW GROVER ACTUALLY WORKS

Grover — Amplitude Amplification



Start
All N states share an equal amplitude of $1/\sqrt{N}$.

Oracle
Flips the SIGN of the marked state — invisible to measurement so far.

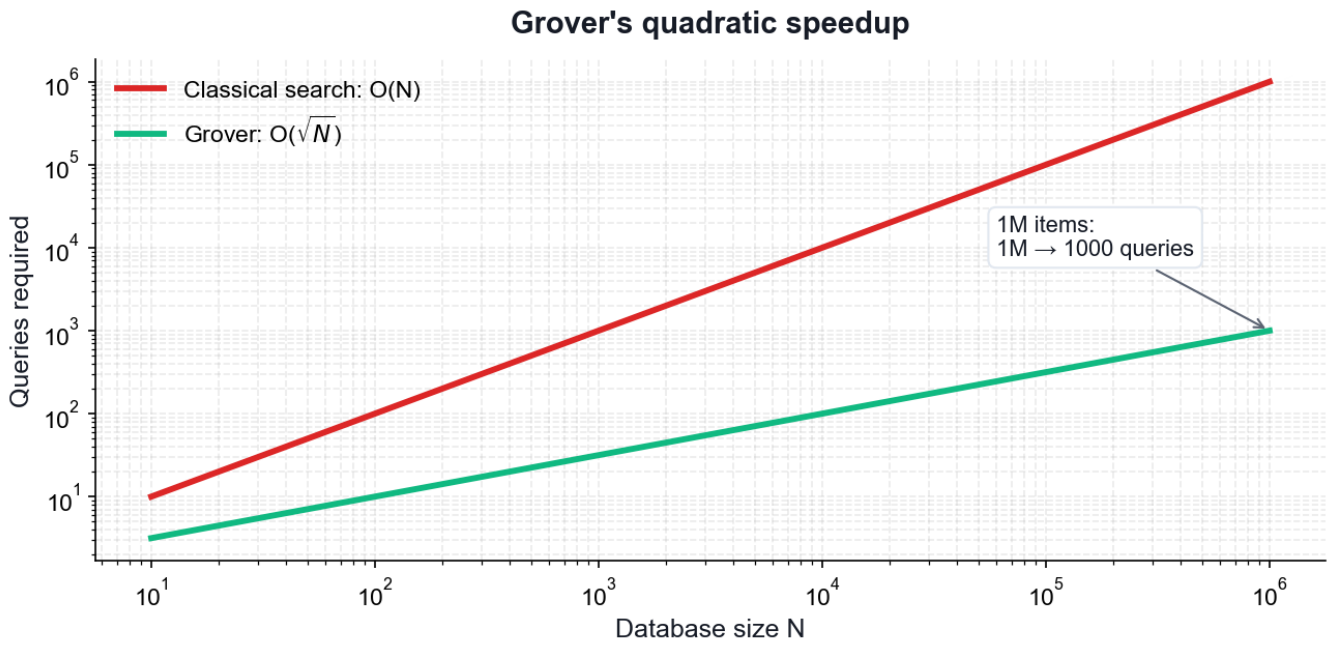
Diffuser
Reflects every amplitude about the mean, growing the marked one.

Repeat
About $(\pi/4)\sqrt{N}$ times — then measure.

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THE SCALING

Grover's Quadratic Speedup



Classical

Checking an unstructured list takes O(N) queries.

Grover

Finds the marked item in O(√N) queries.

1M items

1,000,000 checks become roughly 1,000.

Not exponential

A real speedup, but far smaller than Shor's.

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The problem

GIVEN: a unitary U and one of its eigenvectors |u>, with

$$U|u\rangle = e^{(2\pi i\varphi)}|u\rangle$$

FIND: the phase $\varphi \in [0,1)$, to n bits of precision.

Recall: unitary eigenvalues always have magnitude 1 (Axiom 2), so all the information is in the single real number φ.

KEY POINT: QPE is the workhorse subroutine: order finding, factoring, quantum chemistry energies and HHL all reduce to it.

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Step 1 — Superpose the counting register

Use two registers:
  counting register: t qubits, initialised $|0\rangle^t$
  eigenstate register: holds $|u\rangle$

Apply H to every counting qubit:

$$(1/\sqrt{2^t}) \Sigma_{(k=0..2^t-1)} |k\rangle \ (x) \ |u\rangle$$

KEY POINT: t controls the precision: t bits give φ to t binary places.

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Step 2 — Controlled powers of U (kickback)

Apply controlled-U^(2^j) from counting qubit j onto the eigenstate register.

By PHASE KICKBACK, since |u> is an eigenvector:

C-U^(2^j) |1>|u> = e^(2πi 2^j φ) |1>|u>

Result over the whole register:
(1/√2^t) Σ_k e^(2πi k φ) |k> (x) |u>

KEY POINT: The eigenstate register is UNCHANGED throughout — every bit of the phase has been written onto the counting register. This is phase kickback at scale.

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THE ALGORITHM BEHIND SHOR

Step 3 — Inverse QFT extracts the phase

The counting register now holds $\Sigma_k e^{(2\pi i k\varphi)}|k\rangle$ — exactly the Fourier transform of a peak at φ.

Apply the INVERSE QFT:

$$QFT^{-1}\left(\Sigma_k e^{(2\pi i k\varphi)}|k\rangle\right) \rightarrow |2^t \varphi\rangle$$

Measure the counting register: the result read as a binary fraction is φ.

KEY POINT: If φ has an exact t-bit expansion the answer is exact; otherwise you get the best t-bit approximation with probability ≥ $4/\pi^2 \approx 40\%$.

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Step 4 — Precision and cost

To get φ to n bits with success probability 1-ε:

$$t = n + \lceil \log_2(2 + 1/(2\varepsilon)) \rceil \quad \text{qubits}$$

Cost is dominated by the controlled-U^(2^j) operations.

Worked example — U = Z, |u> = |1>: Z|1> = -|1> = e^(2πi·0.5)|1>
so φ = 0.5, and QPE returns binary 0.1 = 0.5. Verifiable by hand.

KEY POINT: The Z example is the standard classroom check — small enough to trace through by hand, and it exercises the whole pipeline.

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Where QPE is used

ORDER FINDING / SHOR — U: |y> -> |ay mod N>. The phase encodes the period r, so factoring reduces to QPE.

QUANTUM CHEMISTRY — U = e^(-iHt). The phase gives the ENERGY eigenvalue, the fault-tolerant successor to VQE.

HHL (linear systems) — QPE finds the eigenvalues of the matrix A so they can be inverted.

QUANTUM COUNTING — QPE applied to the Grover operator counts solutions.

KEY POINT: If you understand phase kickback and the QFT, you understand the engine inside most of the exponentially-fast quantum algorithms.

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From the classical DFT

The classical Discrete Fourier Transform maps a vector to its spectrum:

$$y_k = (1/\sqrt{N}) \Sigma_{j=0..N-1} x_j \cdot e^{(2\pi i jk/N)}$$

The QFT is the SAME transform, applied to quantum AMPLITUDES:

$$QFT |j\rangle = (1/\sqrt{N}) \Sigma_{k=0..N-1} e^{(2\pi i jk/N)} |k\rangle$$

and by linearity it acts on a whole superposition at once.

KEY POINT: Same mathematics, different substrate: the DFT transforms data in memory, the QFT transforms amplitudes inside a superposition.

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Worked example — QFT on one qubit

For n = 1, N = 2, and e^(2πi/2) = e^(iπ) = -1:

$$QFT|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$$
$$QFT|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$$

That is exactly the HADAMARD gate.

$$QFT_1 = H$$

KEY POINT: The Hadamard is the single-qubit QFT — a useful anchor for intuition.

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The circuit for n qubits

For each qubit, in order:
1. apply H
2. apply controlled-phase rotations R_k from each lower qubit, where R_k adds the phase 2π/2^k

Finally, SWAP to reverse the qubit order.

3-qubit QFT:
H(q2) · CP(π/2: q1->q2) · CP(π/4: q0->q2)
H(q1) · CP(π/2: q0->q1)
H(q0) · SWAP(q0,q2)

KEY POINT: Each controlled rotation adds one further binary digit of phase precision.

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Cost and the honest caveat

QFT gate count: $O(n^2)$ on $N = 2^n$ amplitudes
Classical FFT: $O(N \log N) = O(n \cdot 2^n)$

Exponentially fewer operations — BUT:

You cannot READ OUT the transformed amplitudes. Measuring gives ONE sample, not the spectrum. So the QFT does not accelerate signal processing; it accelerates finding a PERIOD or a PHASE.

KEY POINT: State this caveat explicitly in class — it is the most commonly over-claimed result in quantum computing.

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DERIVATION

Proof — The Deutsch Algorithm

CLAIM

One oracle query decides whether f:{0,1} -> {0,1} is CONSTANT or BALANCED. Any classical algorithm needs two.

GIVEN

f is a black box with f(0), f(1) in {0,1}.

CONSTANT means f(0) = f(1); BALANCED means f(0) ≠ f(1).

The oracle is reversible: $U_f |x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$.

Classically you must learn BOTH f(0) and f(1) — two queries — to compare them.

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Step 1 — Prepare the registers

Start: $|0\rangle\,|1\rangle$

Apply H to both qubits:

$H|0\rangle = |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$

$H|1\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$

State $|\psi_1\rangle = |+\rangle\,|-\rangle$

$= (1/\sqrt{2})(|0\rangle + |1\rangle)\,(x)\,|-\rangle$

WHY THIS STEP: The target is deliberately put in |-> so phase kickback will fire.

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Step 2 — Apply the oracle (phase kickback)

By the kickback identity: $U_f |x\rangle|-\rangle = (-1)^{f(x)} |x\rangle|-\rangle$
Apply it to each term of the superposition:

$|\psi_2\rangle = (1/\sqrt{2})((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle) (x) |-\rangle$

WHY THIS STEP: One query — but the oracle has acted on BOTH inputs at once. The target |-> is unchanged and now factors out entirely.

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Step 3 — Factor out the global phase

Pull (-1)^f(0) out of the bracket:

$|\psi_2\rangle = (-1)^{f(0)} \cdot (1/\sqrt{2})(|0\rangle + (-1)^{(f(0) \oplus f(1))} |1\rangle) (x) |-\rangle$

(-1)^f(0) is a GLOBAL phase — unobservable, so we may drop it.

WHY THIS STEP: Everything now depends on the single bit f(0) ⊕ f(1) — exactly the question we are asking.

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Step 4 — Split into the two cases

CASE A — f is CONSTANT: f(0) = f(1) => f(0) ⊕ f(1) = 0
control = (|0> + |1>)/√2 = |+>

CASE B — f is BALANCED: f(0) ≠ f(1) => f(0) ⊕ f(1) = 1
control = (|0> - |1>)/√2 = |->

WHY THIS STEP: The two cases are ORTHOGONAL states: <+|-> = 0. Orthogonal states can be distinguished perfectly by a single measurement.

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Step 5 — Interfere with a final Hadamard

H is its own inverse and maps the |+>/|-> basis to the |0>/|1> basis:

H|+> = |0>

H|-> = |1>

CASE A (constant): H|+> = |0> -> measure 0 with probability 1

CASE B (balanced): H|-> = |1> -> measure 1 with probability 1

WHY THIS STEP: This is the interference step: a phase difference becomes an amplitude difference.

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Step 6 — Conclusion

Measure the control qubit:

result 0 => f is CONSTANT
result 1 => f is BALANCED

Total oracle queries used: ONE.
Classical worst case: TWO.

WHY THIS STEP: The answer is deterministic — not probabilistic. No repetition is needed.

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QED

The Deutsch Algorithm — Result

RESULT

Measuring 0 proves f is constant; measuring 1 proves f is balanced — with certainty, after a single oracle query.

WHY IT MATTERS

Deutsch was the first proof that a quantum computer can beat every classical algorithm at a well-defined task. It answers a GLOBAL question (is f constant?) without learning the LOCAL values f(0) and f(1) — the template every later algorithm follows.

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DERIVATION

Proof — Deutsch-Jozsa (n qubits)

CLAIM

For f:{0,1}^{n} -> {0,1} PROMISED to be constant or balanced, one query decides which. Classically the worst case is 2^(n-1) + 1 queries.

GIVEN

Promise: f is either constant (same output for all 2^n inputs) or balanced (exactly half the inputs give 0).

Register: n input qubits in $|0\rangle^n$, plus one target qubit in $|1\rangle$.

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Step 1 — Superposition over ALL 2^n inputs

Apply H to each of the n input qubits:

$H^n |0\rangle^n = (1/\sqrt{2^n}) \Sigma_x |x\rangle$ (sum over all x in $\{0,1\}^n$)

Target: $H|1\rangle = |-\rangle$
State: $(1/\sqrt{2^n}) \Sigma_x |x\rangle (x) |-\rangle$

WHY THIS STEP: A single register now holds all 2^n input strings simultaneously.

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Step 2 — Oracle applies the sign

$$U_f |x\rangle|-\rangle = (-1)^{f(x)} |x\rangle|-\rangle$$

State: $(1/\sqrt{2^n}) \Sigma_x (-1)^{f(x)} |x\rangle |(x)\rangle|-\rangle$

WHY THIS STEP: ONE query has tagged every one of the 2^n inputs with the sign of f(x).

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Step 3 — Final Hadamard layer

The n-qubit Hadamard acts as:

$H^n |x\rangle = (1/\sqrt{2^n}) \Sigma_z (-1)^{(x \cdot z)} |z\rangle$ where $x \cdot z = \Sigma x_i z_i \bmod 2$

Applying it:

$(1/2^n) \Sigma_z [\Sigma_x (-1)^{(f(x) + x \cdot z)}] |z\rangle$

WHY THIS STEP: The bracket is the amplitude of outcome |z> — a sum of ±1 terms.

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Step 4 — Amplitude of the all-zeros outcome

Set z = 0…0. Then x·z = 0 for every x, so:

amplitude(|0…0>) = $(1/2^n)\ \Sigma_x\ (-1)^{f(x)}$

CONSTANT: every term is the same sign, so the sum is $\pm 2^n$
amplitude = ±1 -> probability $|\pm 1|^2 = 1$

BALANCED: half the terms are +1 and half are -1, so the sum is 0
amplitude = 0 -> probability 0

WHY THIS STEP: Perfect constructive interference in one case, perfect destructive in the other.

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Step 5 — Conclusion

Measure the n input qubits:

all zeros (000…0) => f is CONSTANT (certainty)
anything else => f is BALANCED (certainty)

Quantum: 1 query. Classical worst case: 2^(n-1) + 1 queries.

WHY THIS STEP: For n = 20 that is 1 query versus 524,289 — an exponential separation.

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QED

Deutsch-Jozsa (n qubits) — Result

RESULT

An all-zeros measurement proves f is constant; any other outcome proves f is balanced. One query, with certainty.

WHY IT MATTERS

Deutsch-Jozsa gives the first EXPONENTIAL separation between quantum and classical query complexity. The pattern — superpose, kick back the phase, interfere — is the skeleton of Bernstein-Vazirani, Simon and ultimately Shor.

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DERIVATION

Proof — Grover's Algorithm and the √N Speedup

CLAIM

Grover finds a marked item among N in O(√N) queries, and (π/4)√N iterations is optimal.

GIVEN

Search space of $N = 2^n$ items with exactly one marked item $|w\rangle$.

Oracle: $O|x\rangle = -|x\rangle$ if $x = w$, else $O|x\rangle = |x\rangle$ (a phase oracle).

Classical search needs O(N) queries — on average N/2.

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Step 1 — Uniform superposition

$$|s\rangle = H^n|0\rangle^n = (1/\sqrt{N}) \Sigma_x |x\rangle$$

Overlap with the marked item: $\langle w|s\rangle = 1/\sqrt{N}$
So P(measuring w now) = 1/N — no better than a random guess.

WHY THIS STEP: Superposition alone gives NO advantage. Interference must do the work.

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Step 2 — Work in a 2-D plane

Define the state orthogonal to the marked item:

$$|s'\rangle = (1/\sqrt{(N-1)}) \Sigma_{(x \neq w)} |x\rangle$$

Then $|s\rangle = \sin(\theta)|w\rangle + \cos(\theta)|s'\rangle$ where $\sin(\theta) = 1/\sqrt{N}$.

Everything that follows stays in the 2-D plane spanned by $|w\rangle$ and $|s'\rangle$.

WHY THIS STEP: Reducing a 2^n-dimensional problem to a 2-D rotation is the key simplification.

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Step 3 — The oracle is a reflection

O flips the sign of the |w> component only:

$$O(\sin\theta|w\rangle + \cos\theta|s'\rangle) = -\sin\theta|w\rangle + \cos\theta|s'\rangle$$

Geometrically: a REFLECTION about the |s'> axis.

WHY THIS STEP: The marked item is now 'marked' by a phase — still invisible to measurement.

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Step 4 — The diffuser is a second reflection

$$D = 2|s\rangle\langle s| - I$$

This reflects any state about |s>. Equivalently: 'invert about the mean'.
Circuit: $H^n \cdot (2|0\rangle\langle 0| - I) \cdot H^n$

WHY THIS STEP: Two reflections in a plane compose to a ROTATION — that is the whole trick.

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Step 5 — Each Grover iteration rotates by 2θ

Composing the two reflections:
G = D·O rotates the state by angle 2θ toward |w>.

After k iterations:
$|\psi_k\rangle = \sin((2k+1)\theta)|w\rangle + \cos((2k+1)\theta)|s'\rangle$

So $P(\text{measure } w) = \sin^2((2k+1)\theta)$.

WHY THIS STEP: The amplitude on |w> grows roughly LINEARLY in k — while probability grows quadratically.

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Step 6 — Optimal number of iterations

Maximise by setting $(2k+1)\theta = \pi/2$:

$$k = \pi/(4\theta) - 1/2$$

For large N, $\sin\theta = 1/\sqrt{N}$ gives $\theta \approx 1/\sqrt{N}$, so:

$$k \approx (\pi/4)\sqrt{N}$$

Example $N = 10^6$: $k \approx 785$ queries versus ~500,000 classically.

WHY THIS STEP: OVER-ROTATION IS REAL: run too many iterations and the state rotates PAST |w>, and the success probability falls again.

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Step 7 — Optimality

Bennett, Bernstein, Brassard and Vazirani (1997) proved that ANY quantum algorithm needs Ω(√N) queries for unstructured search.

Grover therefore achieves the optimal scaling — it cannot be improved.

WHY THIS STEP: The speedup is QUADRATIC, not exponential. This is why Grover only halves effective symmetric key lengths rather than breaking them.

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QED

Grover's Algorithm and the √N Speedup — Result

RESULT

After ≈(π/4)√N iterations the marked state is measured with probability ≈1. The quadratic speedup is provably optimal for unstructured search.

WHY IT MATTERS

Grover is the reason AES-128 drops to ~64-bit effective security against a quantum adversary and why NIST recommends AES-256. Its amplitude-amplification core also underpins quantum counting and amplitude estimation.

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DERIVATION

Proof — The Quantum Fourier Transform is Efficient

CLAIM

The QFT on n qubits needs only $O(n^2)$ gates, versus $O(N \log N) = O(n \cdot 2^n)$ for the classical FFT.

GIVEN

$N = 2^n$. The QFT acts as $|j\rangle \to (1/\sqrt{N}) \sum_k e^{(2\pi i jk/N)} |k\rangle$.

$\omega = e^{(2\pi i/N)}$ is the primitive N-th root of unity.

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Step 1 — The product-form identity

Write k in binary and expand the exponent. The 2^n-term sum FACTORISES:

$$QFT|j\rangle = (1/\sqrt{N}) \otimes_{l=1..n} (|0\rangle + e^{(2\pi i\, j/2^l)} |1\rangle)$$

The n-qubit output is a PRODUCT of n single-qubit states.

WHY THIS STEP: This factorisation is why the QFT is efficient — an exponentially large sum collapses into n independent factors.

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Step 2 — Build each factor: H + CPhase

Qubit l needs the phase $e^{(2\pi i\, j/2^l)}$, assembled from:

H — creates $(|0\rangle + e^{(2\pi i\, 0.j_l)}\,|1\rangle)/\sqrt{2}$
CROT_k — adds the phase $2\pi/2^k$, controlled on a lower qubit

Qubit 1 needs 1 H + (n-1) controlled rotations
Qubit 2 needs 1 H + (n-2) controlled rotations …

WHY THIS STEP: Each controlled rotation contributes one more bit of phase precision.

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Step 3 — Count the gates

Total = n Hadamards + Σ(n-1) + (n-2) + … + 1 controlled rotations

= n + n(n-1)/2

= O(n²)

Plus n/2 SWAPs to reverse the qubit order.

For n = 10: ~55 gates. Classical FFT on N = 1024: ~10,240 operations.

WHY THIS STEP: Quadratic in the number of QUBITS, versus linearithmic in the number of AMPLITUDES — an exponential saving.

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Step 4 — The essential caveat

The QFT transforms AMPLITUDES, which cannot be read out directly. Measuring gives ONE sample from the transformed distribution, not the whole spectrum.

So the QFT is NOT a drop-in replacement for the FFT in signal processing. Its power is in extracting a PERIOD or a PHASE — which is exactly what Shor's algorithm and quantum phase estimation need.

WHY THIS STEP: An honest statement of what the speedup does and does not buy — a common source of hype.

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QED

The Quantum Fourier Transform is Efficient — Result

RESULT

The QFT runs in $O(n^2)$ gates thanks to the product-form factorisation, providing the exponential engine inside phase estimation and Shor's algorithm.

WHY IT MATTERS

The QFT is the single most important subroutine in quantum computing — it is what turns period finding, and therefore factoring, from exponential into polynomial.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Algorithms

Activities 9–10

- Deutsch and Deutsch-Jozsa Algorithms
- Bernstein-Vazirani Algorithm

Activities 11–12

- Grover's Search Algorithm
- Quantum Fourier Transform

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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 9

Deutsch and Deutsch-Jozsa Algorithms

Implement the first algorithms to show a provable quantum advantage: decide whether a black-box function is constant or balanced with a single query, then generalise from one input bit to n.

MAPS TO LO6 — Establish guidelines for selecting quantum computing algorithms

YOU'LL PRODUCE

Deutsch and Deutsch-Jozsa circuits with all four oracle types, run on IBM Quantum hardware, correctly classifying each function in one query.

Tools: Qiskit 2.x, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 9: Deutsch and Deutsch-Jozsa Algorithms

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ACTIVITY 9 · VERIFY & DEBRIEF

Deutsch and Deutsch-Jozsa Algorithms

TEST IT — HOW YOU KNOW IT WORKED

The Deutsch circuit measures 0 for both constant oracles and 1 for both balanced oracles; Deutsch-Jozsa returns the all-zeros string for a constant function and a non-zero string for a balanced function — each using exactly one oracle query.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 10

Bernstein-Vazirani Algorithm

Recover an n-bit hidden string in a single oracle query. Classically you would need n queries — one per bit — making this a clean demonstration of linear-to-constant query speedup.

MAPS TO LO6 — Establish guidelines for selecting quantum computing algorithms

YOU'LL PRODUCE

A Bernstein-Vazirani circuit that recovers a chosen hidden string such as 1011 in one query on real hardware.

Tools: Qiskit 2.x, IBM Quantum Runtime

Full step-by-step guide: Learner Guide — Activity 10: Bernstein-Vazirani Algorithm

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ACTIVITY 10 · VERIFY & DEBRIEF

Bernstein-Vazirani Algorithm

TEST IT — HOW YOU KNOW IT WORKED

The most frequent measured bit string equals the hidden string s that was encoded in the oracle, recovered with a single oracle query regardless of the length of s.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 11

Grover's Search Algorithm

Implement amplitude amplification to find a marked item in an unstructured search space, using the phase oracle and the diffuser, and see the quadratic O(sqrt(N)) speedup for 2-qubit and 3-qubit searches.

MAPS TO LO6, LO7 — Select algorithms and synthesize quantum methodologies

YOU'LL PRODUCE

Grover circuits for 2 and 3 qubits that amplify the marked state to a high measurement probability on real hardware.

Tools: Qiskit 2.x, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 11: Grover's Search Algorithm

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ACTIVITY 11 · VERIFY & DEBRIEF

Grover's Search Algorithm

TEST IT — HOW YOU KNOW IT WORKED

For 2 qubits the marked state is measured with near-certainty after one Grover iteration; for 3 qubits it dominates the histogram after two iterations; applying too many iterations visibly reduces the success probability.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 05 · HANDS-ON ACTIVITY

ACTIVITY 12

Quantum Fourier Transform

Build the QFT circuit from Hadamard and controlled-phase gates, verify it against the classical DFT, and understand why it is the engine behind phase estimation and Shor's algorithm.

MAPS TO LO6, LO7 — Establish guidelines for selecting quantum computing algorithms

YOU'LL PRODUCE

QFT circuits for 3, 4 and 5 qubits, validated against a classical DFT reference and run on IBM Quantum hardware.

Tools: Qiskit 2.x, NumPy, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 12: Quantum Fourier Transform

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ACTIVITY 12 · VERIFY & DEBRIEF

Quantum Fourier Transform

TEST IT — HOW YOU KNOW IT WORKED

Your hand-built QFT circuit produces the same state vector as Qiskit's built-in QFT and matches a normalised classical FFT of the same input, using on the order of n^2 gates.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Algorithms

- The query (oracle) model
- Deutsch algorithm
- Deutsch-Jozsa
- Bernstein-Vazirani
- Grover search
- Quantum Fourier Transform

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06

TOPIC 06

Quantum Cryptography and Quantum Safety

Classical Cryptography · The Quantum Threat · Shor Algorithm · Post-Quantum Cryptography

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MAPS TO LO4, LO7

Key Concepts — Quantum Cryptography and Quantum Safety

1

Goals of cryptography

Confidentiality, integrity, authentication and non-repudiation are the four properties every scheme must address.

2

Classical schemes

Symmetric (AES), asymmetric (RSA, ECC) and hash functions (SHA-256) rest on problems that are hard for classical computers.

3

The quantum threat

Shor's algorithm breaks RSA and ECC in polynomial time; Grover halves the effective key length of symmetric ciphers.

4

Shor's algorithm

Reduces factoring to period-finding, then uses the QFT to extract the period — turning a hard problem into an easy one.

5

Harvest now, decrypt later

Adversaries can record encrypted traffic today and decrypt it once a cryptographically relevant quantum computer exists.

6

Post-quantum cryptography

NIST has standardised ML-KEM (Kyber), ML-DSA (Dilithium), SLH-DSA (SPHINCS+) and HQC as quantum-resistant replacements.

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WHAT BREAKS AND WHAT SURVIVES

The Quantum Threat to Cryptography

The Quantum Threat to Today's Cryptography

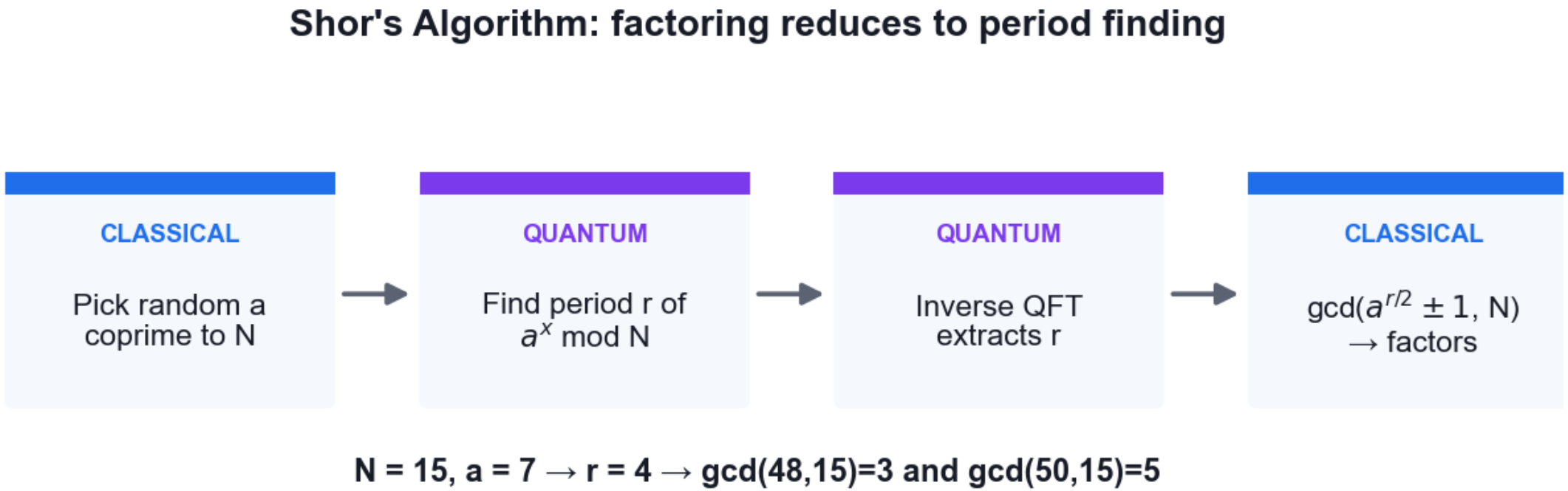
BROKEN by Shor	WEAKENED by Grover	QUANTUM-SAFE (NIST)
RSA-2048	AES-128 → 64-bit	ML-KEM (FIPS 203)
ECC / ECDSA	SHA-256 → 128-bit	ML-DSA (FIPS 204)
Diffie-Hellman	AES-256 → safe	SLH-DSA (FIPS 205)
DSA	SHA-512 → safe	HQC (2025)
Polynomial time — public-key crypto fails	Effective key length is halved — double it	Standardised replacements

Harvest now, decrypt later — data captured today is at risk the moment a CRQC arrives

Symmetric crypto survives by doubling key length; public-key crypto must be replaced outright.

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Shor's Algorithm



Only the period-finding step is quantum — the rest is classical number theory.

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Classical Cryptography — What You Must Know

Type	Example	Key	Broken by
Symmetric	AES-256	One shared secret key	Grover (halves strength)
Asymmetric	RSA-2048, ECC	Public + private key pair	Shor (fully broken)
Hash	SHA-256	No key	Grover (halves strength)
PQC KEM	ML-KEM (Kyber)	Lattice-based	Not known to be broken
PQC signature	ML-DSA, SLH-DSA	Lattice / hash-based	Not known to be broken

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WHY MIGRATE BEFORE THE THREAT ARRIVES

Harvest now, decrypt later.

An adversary can record your encrypted traffic today and decrypt it the day a cryptographically relevant quantum computer exists. Any secret that must stay confidential for a decade is already at risk.

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WHAT TO DO ON MONDAY MORNING

The threat timeline

Three dates decide your risk (Mosca's theorem):

X = how long your data must stay secret (years)
Y = how long migration will take you (years)
Z = time until a cryptographically relevant QC (years)

IF X + Y > Z then you are ALREADY exposed.

Health records X=25y, migration Y=5y, Z estimated 10-15y -> act now.

KEY POINT: Mosca's inequality turns a vague fear into a concrete deadline you can put in front of a board.

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WHAT TO DO ON MONDAY MORNING

Step 1 — Build a cryptographic inventory (CBOM)

You cannot migrate what you cannot see. Enumerate:

- TLS certificates and their key algorithms
- VPNs, SSH keys, code-signing certificates
- database and backup encryption
- hardcoded crypto in applications and firmware
- third-party and SaaS dependencies

Produce a Cryptographic Bill of Materials (CBOM). Tools: CBOM/CycloneDX, sslyze, openssl scanning, source-code scanners.

KEY POINT: Most organisations discover far more RSA and ECC than they expected, especially embedded in vendor products.

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Step 2 — Classify and prioritise

```
BROKEN by Shor      RSA, ECC, ECDSA, ECDH, DH, DSA   ->  must replace
WEAKENED by Grover  AES-128, SHA-256                 ->  double key size
SAFE                AES-256, SHA-512, SHA-3          ->  keep
```

Priority score = (data lifetime) × (exposure) × (replacement difficulty)

Rank highest: long-lived secrets, long-lived devices, anything whose traffic crosses an untrusted network today.

KEY POINT: Harvest-now-decrypt-later means long confidentiality lifetimes dominate the ranking — not how sensitive the system feels today.

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Step 3 — Deploy the NIST standards

FIPS 203 ML-KEM (Kyber) key encapsulation — replaces RSA/ECDH
FIPS 204 ML-DSA (Dilithium) signatures — the general-purpose default
FIPS 205 SLH-DSA (SPHINCS+) hash-based signatures — conservative backup
2025 HQC code-based KEM — algorithmic diversity

Chosen for structural DIVERSITY: lattice, hash and code-based problems, so one mathematical breakthrough cannot break everything at once.

KEY POINT: SLH-DSA has larger signatures but rests only on hash-function security — the most conservative assumption available.

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Step 4 — Hybrid mode during transition

Run classical AND post-quantum together, combining both shared secrets:

```
shared_secret = KDF( ECDH_secret || ML-KEM_secret )
```

The session stays secure if EITHER component holds.

Already deployed: X25519+ML-KEM768 in Chrome, Firefox, Cloudflare, Signal (PQXDH), Apple iMessage (PQ3), OpenSSH 9.x.

KEY POINT: Hybrid protects against both a future quantum attack AND an implementation flaw in the new, less battle-tested PQC code.

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WHAT TO DO ON MONDAY MORNING

Step 5 — Build crypto-agility

The real lesson of this migration is that there will be ANOTHER one.

- never hardcode algorithms — make them configuration
- negotiate algorithms at the protocol level
- centralise crypto in a library or service, not scattered in apps
- keep certificate lifetimes short so rotation is routine
- test algorithm swaps in CI

Design goal: changing algorithm should be a CONFIG change, not a rewrite.

KEY POINT: Organisations that built crypto-agility for the SHA-1 to SHA-2 migration are finding the PQC transition dramatically cheaper.

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Two meanings of 'quantum security'

1. SECURITY AGAINST quantum computers
   = Post-Quantum Cryptography (PQC)
   Classical maths, classical hardware, believed quantum-resistant.
   Runs on the machines you already own.

2. SECURITY USING quantum physics
   = QKD, QRNG, device-independent cryptography
   Needs quantum hardware; security from physical law.

KEY POINT: These are constantly conflated in vendor marketing. Know which one is being sold to you.

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The quantum threat model

SHOR breaks RSA, DH, ECC, ECDSA outright — polynomial time.
All public-key crypto in use today fails.

GROVER halves effective symmetric key length.
AES-128 -> ~64-bit security; AES-256 stays safe.
(In practice the parallelisation penalty makes this weaker still.)

HARVEST NOW, DECRYPT LATER
The attack is happening TODAY. Encrypted traffic is being recorded now for decryption once a CRQC exists.

KEY POINT: Note the asymmetry: symmetric crypto survives with bigger keys; public-key crypto must be replaced entirely.

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QKD vs PQC — an honest comparison

QKD	PQC
security from physics	security from hard maths
needs dedicated fibre/satellite	runs on existing hardware
~100-500 km without repeaters	unlimited range (internet)
expensive dedicated hardware	software update
no authentication built in	signatures included
point-to-point links	works at internet scale

NSA/NCSC advise PQC for general use; QKD suits specific high-value links.

KEY POINT: QKD is NOT a drop-in replacement for public-key infrastructure — it still needs an authenticated classical channel to prevent man-in-the-middle.

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Quantum random number generation (QRNG)

Classical PRNGs are DETERMINISTIC — given the seed, the output follows.

QRNG derives randomness from a genuinely indeterministic quantum event:

- which path a photon takes at a beamsplitter
- vacuum fluctuation amplitude
- arrival-time jitter of photons

Axiom 4 (the Born rule) guarantees the outcome is fundamentally random, not merely hard to predict.

KEY POINT: QRNG is the most mature and deployable quantum security technology today — and weak randomness is a very common real-world failure.

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Defence-in-depth for the quantum era

NOW inventory crypto; switch to AES-256 and SHA-384/512;
deploy hybrid key exchange; buy crypto-agility

NEXT migrate signatures to ML-DSA; PQC certificate chains;
update embedded and IoT firmware pipelines

ONGOING monitor NIST/vendor guidance; re-run the CBOM;
treat algorithm agility as a permanent capability

SPECIAL QKD only where the threat model and budget genuinely justify it

KEY POINT: The sequencing matters: confidentiality (harvest-now risk) before authenticity (which an attacker must break in real time).

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DERIVATION

Proof — Shor's Algorithm Breaks RSA

CLAIM

Factoring reduces to period finding, which a quantum computer solves in polynomial time.

GIVEN

Goal: factor a composite N (e.g. an RSA modulus).

Best known classical method (GNFS) is sub-exponential: exp(O((log N)^(1/3))).

Shor runs in $O((\log N)^3)$ — polynomial.

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Step 1 — Reduce factoring to order finding

Pick a random a with 1 < a < N.
If gcd(a, N) ≠ 1 we already have a factor (lucky — done).

Otherwise find the ORDER r: the smallest r > 0 with
$a^r \equiv 1 \pmod N$

WHY THIS STEP: This reduction is entirely CLASSICAL number theory — no quantum yet.

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Step 2 — Why the order gives a factor

If r is EVEN, then $a^r - 1 \equiv 0 \pmod N$ factors as:

$$(a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod N$$

So N divides that product. Provided $a^{r/2} \not\equiv -1 \pmod N$,
$\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$
are NON-TRIVIAL factors of N.

P(r even AND the condition holds) ≥ 1/2 for random a — so a few tries suffice.

WHY THIS STEP: gcd is computed classically in polynomial time by Euclid's algorithm.

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Step 3 — Worked example: N = 15, a = 7

Powers of 7 mod 15:
$7^1 = 7,\ 7^2 = 49 \equiv 4,\ 7^3 \equiv 28 \equiv 13,\ 7^4 \equiv 91 \equiv 1$

So r = 4 (even, good).
a^(r/2) = $7^2 = 49 \equiv 4 \pmod{15}$, and $4 \not\equiv -1$, so we may proceed.

gcd(4 - 1, 15) = gcd(3, 15) = 3
gcd(4 + 1, 15) = gcd(5, 15) = 5

15 = 3 × 5 FACTORED

WHY THIS STEP: Everything here was classical — the only hard step is FINDING r.

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Step 4 — The quantum part: period finding

Prepare a counting register in superposition and compute a^x mod N:

$$(1/\sqrt{Q})\ \Sigma_x\ |x\rangle\ |a^x \bmod N\rangle$$

Measure the second register. It collapses to some value $a^{(x_0)} \bmod N$, leaving the first register in a superposition of all x with the same value:

$$|x_0\rangle + |x_0 + r\rangle + |x_0 + 2r\rangle + \ldots$$

— a PERIODIC state with period exactly r.

WHY THIS STEP: The period is encoded in the register, but measuring directly would just give one random x — useless. The QFT is needed to expose the period.

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Step 5 — The QFT extracts the period

Apply the inverse QFT to the counting register.

The QFT of a periodic comb is a comb with reciprocal spacing: the amplitude concentrates on multiples of Q/r. Measuring yields, with high probability,

$$y \approx k \cdot Q/r \qquad \text{for some integer } k$$

Then $y/Q \approx k/r$, and CONTINUED FRACTIONS recover r from that rational approximation.

WHY THIS STEP: The QFT costs only $O(n^2)$ gates — this is where the exponential speedup lives.

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Step 6 — Cost and consequence

Total: $O((\log N)^3)$ operations — POLYNOMIAL in the number of bits.

RSA-2048 estimates: roughly 4,000 logical qubits and ~10^9 Toffoli gates, which at realistic error rates means MILLIONS of physical qubits.

Shor also breaks Diffie-Hellman and ECC (the discrete-log problem).

WHY THIS STEP: Not achievable today — but 'harvest now, decrypt later' means data captured today is already at risk.

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QED

Shor's Algorithm Breaks RSA — Result

RESULT

Given the order r of a mod N, the factors follow classically from gcd(a^(r/2) ± 1, N). The quantum computer supplies r in polynomial time via the QFT.

WHY IT MATTERS

Shor is the reason post-quantum cryptography exists. Every public-key system in use today — RSA, DH, ECDSA — is broken by it, which is why NIST standardised ML-KEM, ML-DSA and SLH-DSA.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Cryptography and Quantum Safety

Activity 13

- Shor's Algorithm and the Quantum Threat to RSA

Activity 14

- Post-Quantum Cryptography Migration Assessment

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TOPIC 06 · HANDS-ON ACTIVITY

ACTIVITY 13

Shor's Algorithm and the Quantum Threat to RSA

See how factoring reduces to period finding, run the quantum period-finding circuit that sits at the heart of Shor's algorithm, and recover the factors of N = 15 from the measured period.

MAPS TO LO7 — Synthesize quantum methodologies to address complex problems

YOU'LL PRODUCE

A period-finding circuit whose measured output yields the period r, from which the factors of N = 15 are classically recovered.

Tools: Qiskit 2.x, IBM Quantum Runtime, Python, OpenQASM

Full step-by-step guide: Learner Guide — Activity 13: Shor's Algorithm and the Quantum Threat to RSA

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ACTIVITY 13 · VERIFY & DEBRIEF

Shor's Algorithm and the Quantum Threat to RSA

TEST IT — HOW YOU KNOW IT WORKED

The measured phase yields period r = 4 for a = 7 and N = 15, and the classical post-processing step recovers the correct factors 3 and 5.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 06 · HANDS-ON ACTIVITY

ACTIVITY 14

Post-Quantum Cryptography Migration Assessment

Assess which cryptography in a realistic system is broken by a quantum computer, map each algorithm to its NIST post-quantum replacement, and produce a prioritised migration plan against the harvest-now-decrypt-later risk.

MAPS TO LO4, LO7 — Evaluate quantum applicability in the business context

YOU'LL PRODUCE

A completed cryptographic inventory and a prioritised PQC migration plan for a sample organisation.

Tools: NIST FIPS 203/204/205, OpenSSL, Python

Full step-by-step guide: Learner Guide — Activity 14: Post-Quantum Cryptography Migration Assessment

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ACTIVITY 14 · VERIFY & DEBRIEF

Post-Quantum Cryptography Migration Assessment

TEST IT — HOW YOU KNOW IT WORKED

Your inventory correctly separates Shor-broken asymmetric algorithms from Grover-weakened symmetric ones, maps each to the correct NIST FIPS standard, and ranks migration priority by how long the data must remain confidential.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Cryptography and Quantum Safety

- Goals of cryptography
- Classical schemes
- The quantum threat
- Shor's algorithm
- Harvest now, decrypt later
- Post-quantum cryptography

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07

TOPIC 07

Quantum Computing Applications and Use Cases

Finance · Healthcare · Science

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MAPS TO LO4

Key Concepts — Quantum Computing Applications and Use Cases

1

Finance

Portfolio optimisation, derivative pricing via amplitude estimation, credit risk and fraud detection are the leading finance use cases.

2

Healthcare and life sciences

Molecular simulation with VQE, protein folding and accelerated drug discovery target problems classical HPC cannot reach.

3

Science and materials

Simulating catalysts, superconductors, nitrogen fixation and battery chemistry is quantum computing's most natural application.

4

Optimisation and logistics

QAOA addresses routing, scheduling and Max-Cut style combinatorial problems across supply chains and energy grids.

5

Quantum machine learning

Quantum kernels and variational classifiers explore feature spaces that are expensive to compute classically.

6

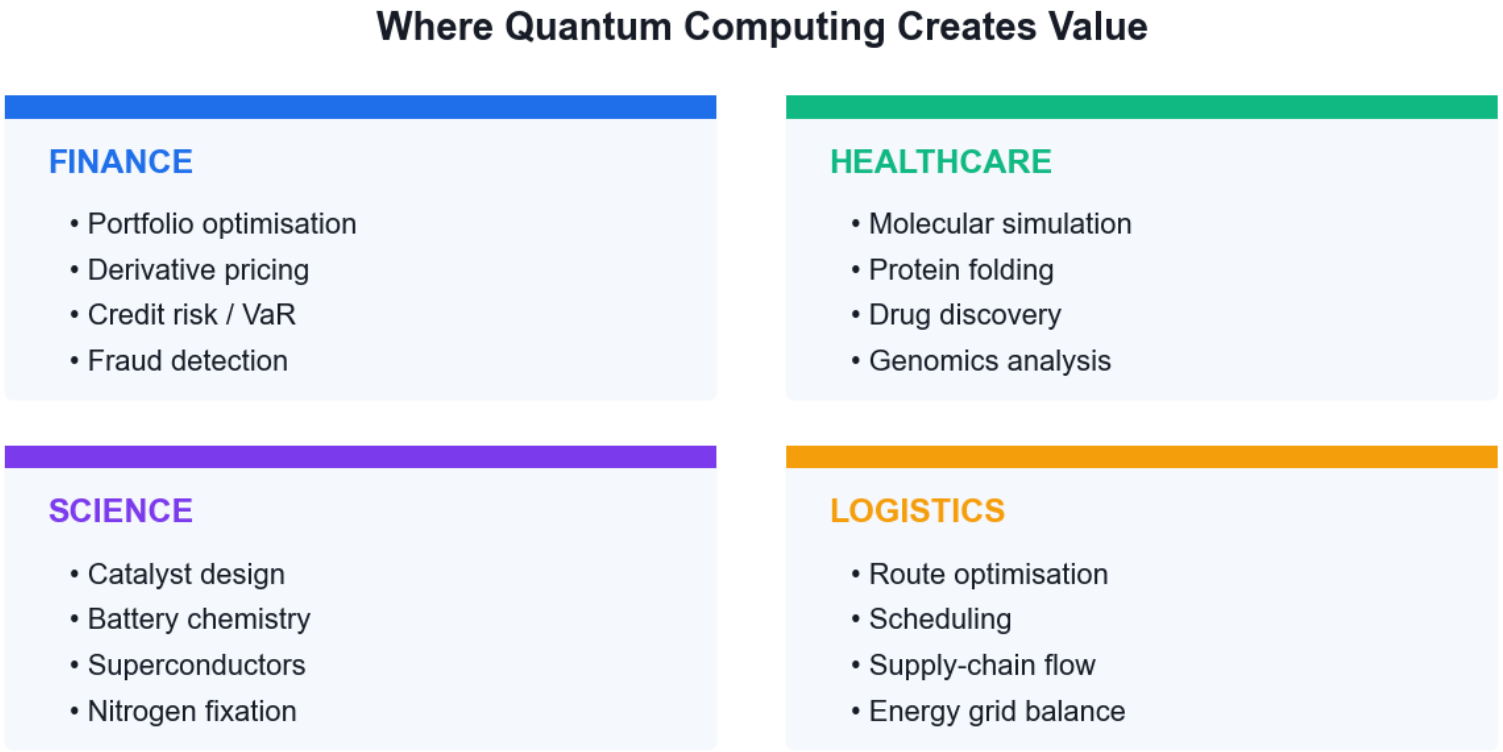
Assessing business value

Evaluate a use case on problem structure, data size, required qubit count and whether a classical method already suffices.

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USE CASES BY SECTOR

Where Quantum Creates Value

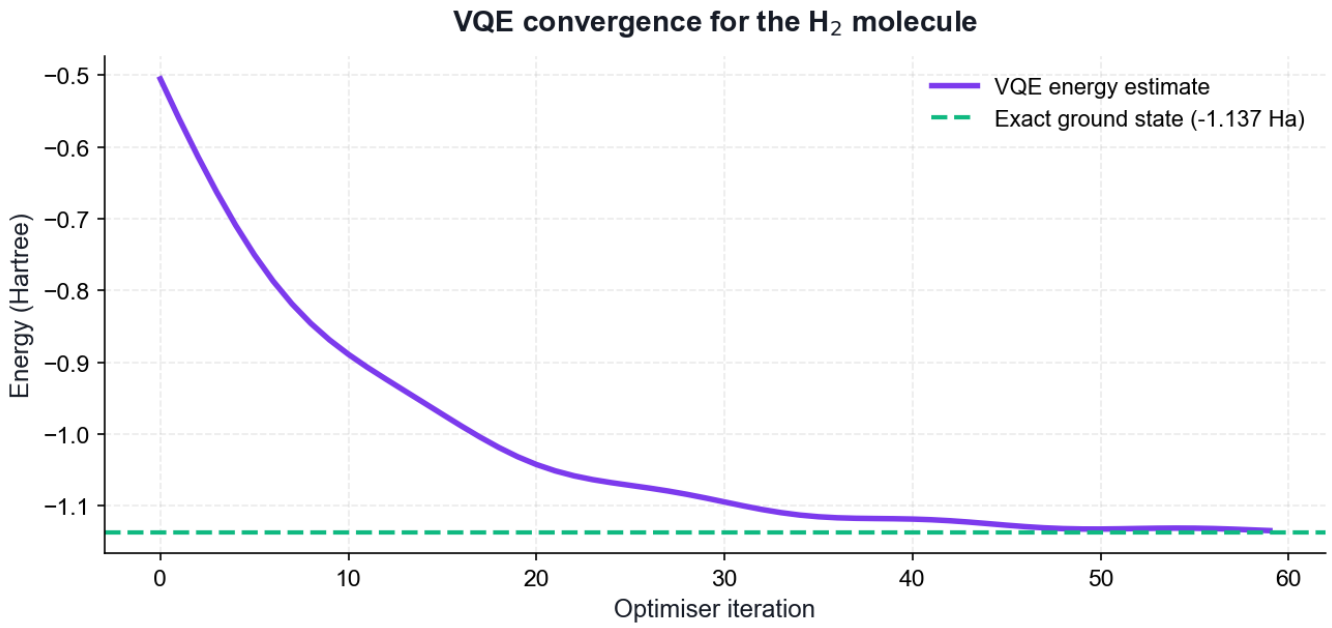


The best candidates are problems with quantum structure — chemistry and optimisation — not big-data problems.

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THE FLAGSHIP NEAR-TERM APPLICATION

VQE — Simulating Molecules



Hybrid loop

A quantum circuit prepares the trial state; a classical optimiser tunes it.

Noise tolerant

Shallow circuits make VQE practical on today's NISQ hardware.

The target

Ground-state energy — the quantity that drives chemistry.

The prize

Catalysts, batteries and drug candidates simulated exactly.

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A PRACTICAL CHECKLIST FOR LO4

Assessing a Quantum Use Case

1

Does the problem have quantum structure?

Chemistry and optimisation map naturally; generic data processing does not.

2

How big is the data?

Loading large classical datasets into a quantum computer is itself a bottleneck.

3

What qubit count is needed?

Estimate logical qubits, then multiply for error-correction overhead.

4

Is there a good classical method?

If a classical heuristic already solves it well, quantum adds nothing.

5

What is the time horizon?

NISQ demos today; fault-tolerant advantage is a 2030s proposition.

6

What is the cost of waiting?

For cryptography the answer is high — migrate now.

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WHERE THE TWO TECHNOLOGIES MEET

Four distinct intersections

1. QUANTUM FOR AI — quantum algorithms to speed up machine learning
2. AI FOR QUANTUM — machine learning to make quantum computers work
3. QUANTUM-INSPIRED — classical algorithms borrowed from quantum ideas
4. SHARED INFRASTRUCTURE — hybrid classical/quantum/GPU pipelines

Intersection 2 is where the REAL commercial value sits today —
which is the reverse of what most headlines suggest.

KEY POINT: Be precise about which intersection is being discussed; they have completely different maturity levels.

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WHERE THE TWO TECHNOLOGIES MEET

Quantum machine learning — the honest state

QUANTUM KERNELS map data into a Hilbert space and compute inner products a classical kernel cannot easily reach

VARIATIONAL (VQC) parameterised circuits trained like a neural net

QUANTUM NEURAL NETS layered ansätze with gradient-based training

REALITY CHECK:

- data loading is the bottleneck — encoding N classical values can cost O(N), destroying any speedup
- BARREN PLATEAUS: gradients vanish exponentially with qubit count
- no proven advantage on a real-world ML dataset yet

KEY POINT: Teach this honestly. Quantum ML is a genuine research field with real open problems, not a shipping product.

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WHERE THE TWO TECHNOLOGIES MEET

AI for quantum — the part that works today

Machine learning is already ESSENTIAL to running quantum hardware:

- QUBIT CALIBRATION — RL agents tune pulse parameters continuously
- ERROR DECODING — neural decoders beat classical matching on surface codes, and run fast enough for real-time correction
- CIRCUIT COMPILATION — ML finds better gate decompositions and routing
- NOISE CHARACTERISATION — learn the device error model, then mitigate
- ANSATZ DESIGN — search the space of variational circuits

Google, IBM and Quantinuum all use ML decoders in production stacks.

KEY POINT: This is the intersection with proven, deployed value — and a strong career area for someone with both skill sets.

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WHERE THE TWO TECHNOLOGIES MEET

Quantum-inspired classical algorithms

Studying quantum algorithms has produced better CLASSICAL ones:

- TENSOR NETWORKS (MPS, PEPS) — from quantum many-body physics, now used to compress neural networks
- Ewin Tang's 'dequantisation' — classical algorithms matching the quantum recommendation-system speedup
- simulated annealing variants inspired by quantum annealing

A quantum algorithm that gets dequantised is still a scientific win — you end up with a faster classical method.

KEY POINT: Useful framing for business stakeholders: quantum R&D pays dividends even when the quantum hardware is not ready.

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WHERE THE TWO TECHNOLOGIES MEET

Where they converge: chemistry

The most credible near-term combination:

1. QUANTUM computer simulates a molecule that is classically intractable
2. Results become high-quality TRAINING DATA
3. An AI MODEL (e.g. a graph neural net) learns from that data
4. The AI screens MILLIONS of candidate molecules cheaply
5. The quantum computer verifies only the top candidates

Quantum supplies scarce, high-accuracy ground truth; AI supplies scale.

KEY POINT: This division of labour plays to each technology's actual strength instead of pitting them against each other.

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WHERE THE TWO TECHNOLOGIES MEET

Security implications of the convergence

AI ACCELERATES CRYPTANALYSIS — ML-assisted side-channel attacks and lattice-reduction heuristics are active research areas

PQC MUST PROTECT AI SYSTEMS — model weights, training data and inference APIs all need quantum-safe transport and signing

AI DEFENDS — anomaly detection can flag harvest-now traffic capture

MODEL PROVENANCE — signing model artifacts with ML-DSA guards the AI supply chain against tampering

KEY POINT: Model weights are long-lived, extremely high-value secrets — a textbook harvest-now-decrypt-later target.

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Realistic outlook

TODAY (NISQ) AI runs quantum hardware. Quantum ML is research.
~2029 (early FT) first logical qubits; quantum chemistry feeds AI training sets; PQC migration should be complete
2030s fault-tolerant chemistry and materials at scale; quantum-generated data becomes routine in AI pipelines

CONSTANT the security work does NOT wait for any of this. Harvest-now-decrypt-later is a TODAY problem.

KEY POINT: The security timeline is independent of the computing timeline — that is the single most important message to take back to work.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Computing Applications and Use Cases

Activity 15

- Molecular Simulation with VQE (Healthcare & Science)

Activity 16

- Portfolio Optimisation with QAOA (Finance)

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TOPIC 07 · HANDS-ON ACTIVITY

ACTIVITY 15

Molecular Simulation with VQE (Healthcare & Science)

Run the Variational Quantum Eigensolver to compute the ground-state energy of the hydrogen molecule — the flagship near-term use case for drug discovery, catalysis and materials science.

MAPS TO LO4, LO7 — Evaluate quantum computing applicability in the business context

YOU'LL PRODUCE

A VQE run that converges to the H2 ground-state energy, with an energy-versus-iteration convergence plot.

Tools: Qiskit 2.x, SciPy optimiser, matplotlib, IBM Quantum

Full step-by-step guide: Learner Guide — Activity 15: Molecular Simulation with VQE (Healthcare & Science)

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ACTIVITY 15 · VERIFY & DEBRIEF

Molecular Simulation with VQE (Healthcare & Science)

TEST IT — HOW YOU KNOW IT WORKED

The VQE optimisation converges to an energy close to the exact H2 ground-state value of about -1.137 Hartree, and the convergence plot shows the energy decreasing monotonically toward that limit.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 07 · HANDS-ON ACTIVITY

ACTIVITY 16

Portfolio Optimisation with QAOA (Finance)

Apply the Quantum Approximate Optimization Algorithm to a Max-Cut style combinatorial problem, the same structure that underlies portfolio selection, routing and scheduling in financial services.

MAPS TO LO4, LO6 — Evaluate applicability and select appropriate algorithms

YOU'LL PRODUCE

A QAOA run that finds the optimal cut on a small graph, with the solution quality compared against brute force.

Tools: Qiskit 2.x, NetworkX, SciPy, matplotlib

Full step-by-step guide: Learner Guide — Activity 16: Portfolio Optimisation with QAOA (Finance)

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ACTIVITY 16 · VERIFY & DEBRIEF

Portfolio Optimisation with QAOA (Finance)

TEST IT — HOW YOU KNOW IT WORKED

The most frequently sampled bit string from the optimised QAOA circuit corresponds to the true maximum cut found by brute force, and the approximation ratio improves as the number of layers p increases.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Computing Applications and Use Cases

- Finance
- Healthcare and life sciences
- Science and materials
- Optimisation and logistics
- Quantum machine learning
- Assessing business value

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08

TOPIC 08

Quantum Error Correction

Bit and Phase Flip Errors · QEC Algorithms · Fault Tolerance · The Future

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MAPS TO LO7

Key Concepts — Quantum Error Correction

1

Why QEC is needed

Qubits decohere and gates are imperfect; without correction, errors accumulate long before a useful algorithm completes.

2

Quantum-specific challenges

No-cloning forbids simple copying, errors are continuous rather than discrete, and measurement destroys the state.

3

Syndrome measurement

Ancilla qubits are measured to reveal WHICH error occurred without ever measuring — and collapsing — the logical state.

4

Bit-flip and phase-flip codes

Three-qubit repetition codes correct X errors; conjugating with Hadamards turns phase flips into bit flips.

5

Shor, Steane and surface codes

Shor's 9-qubit code corrects arbitrary single-qubit errors; Steane uses 7; surface codes are today's leading practical approach.

6

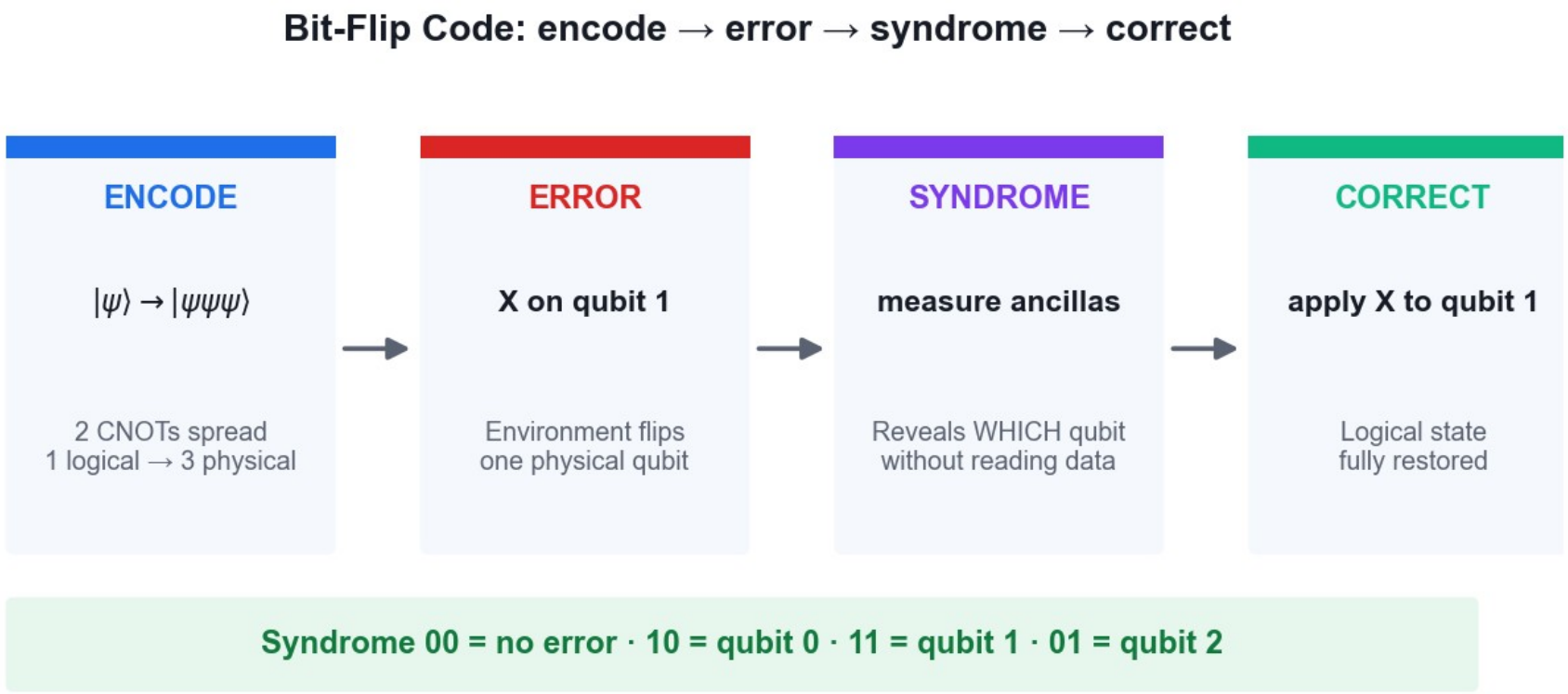
Fault tolerance

Below the error threshold, concatenating codes suppresses errors faster than they accumulate — the path to useful quantum computers.

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ENCODE · ERROR · SYNDROME · CORRECT

Bit-Flip Code — The QEC Cycle

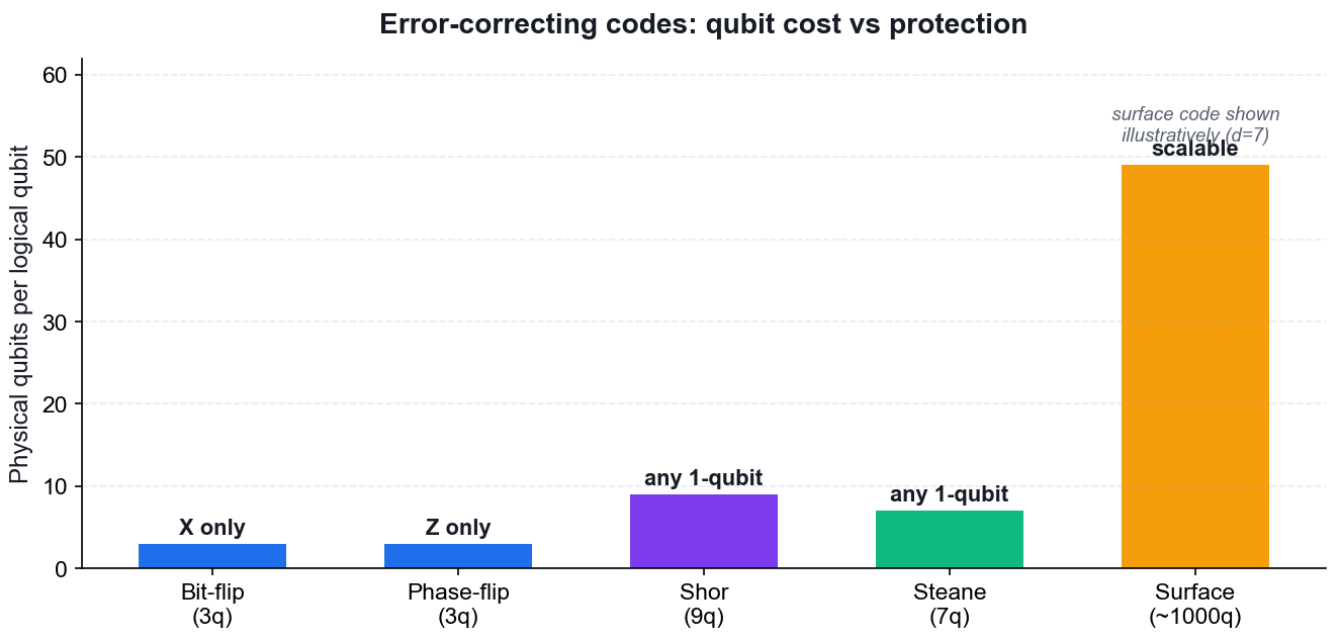


The syndrome reveals WHICH qubit flipped without ever measuring — and destroying — the logical state.

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QUBIT COST VS PROTECTION

Error-Correcting Codes Compared



3-qubit codes

Correct either bit flips or phase flips, but not both.

Shor 9-qubit

The first code to correct an arbitrary single-qubit error.

Steane 7-qubit

A CSS code achieving the same protection more efficiently.

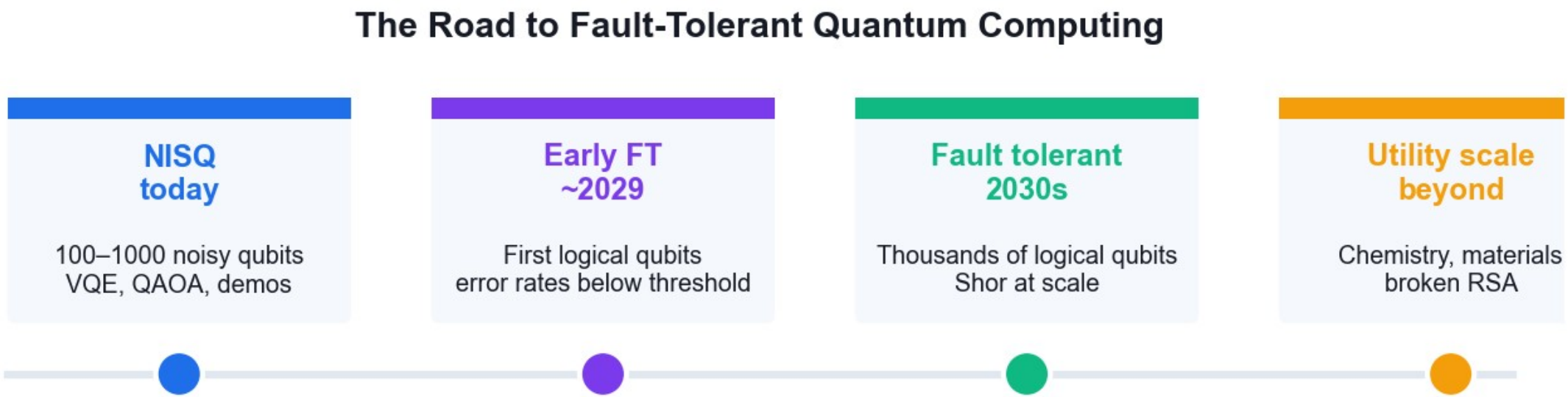
Surface codes

2D nearest-neighbour layout — today's leading practical approach.

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WHERE THE INDUSTRY IS HEADED

The Road to Fault Tolerance



Below the error threshold, adding more physical qubits suppresses errors faster than they accumulate.

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DERIVATION

Proof — Quantum Error Correction Works

CLAIM

Syndrome measurement identifies WHICH error occurred without learning — and therefore without collapsing — the encoded logical state.

GIVEN

Encode one logical qubit into three physical qubits (the bit-flip code).

Logical states: $|0\rangle_L = |000\rangle$, $|1\rangle_L = |111\rangle$.

An arbitrary logical state: $|\psi\rangle_L = a|000\rangle + b|111\rangle$.

Assume at most ONE qubit suffers an X (bit-flip) error.

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Step 1 — Why naive copying is forbidden

The obvious classical fix is to make three copies:
$|\psi\rangle|\psi\rangle|\psi\rangle = (a|0\rangle+b|1\rangle)(a|0\rangle+b|1\rangle)(a|0\rangle+b|1\rangle)$

But no-cloning FORBIDS this for unknown $|\psi\rangle$.

What we build instead is the ENTANGLED state
$a|000\rangle + b|111\rangle$ which is NOT $|\psi\rangle|\psi\rangle|\psi\rangle$

WHY THIS STEP: Encoding spreads the information across three qubits without ever duplicating it.

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Step 2 — Encoding circuit

Two CNOTs, both controlled by the data qubit:

```
  (a|0> + b|1>)|0>|0>
    --CNOT(0->1)-->   a|00>|0> + b|11>|0>
    --CNOT(0->2)-->   a|000>   + b|111>

= |ψ>_L
```

WHY THIS STEP: Note a and b are never measured, never learned, never copied.

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Step 3 — An error strikes

Suppose X acts on qubit 1:

$$X_1 (a|000\rangle + b|111\rangle) = a|010\rangle + b|101\rangle$$

The state is corrupted — but observe that the amplitudes a and b are INTACT.
Only the BASIS LABELS have been permuted.

WHY THIS STEP: This is the crucial observation: the error moved the state to a different, distinguishable subspace without touching the encoded information.

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Step 4 — Syndrome operators

Measure the two PARITY observables (stabilisers):

$S_1 = Z_0Z_1$ (parity of qubits 0 and 1)
$S_2 = Z_1Z_2$ (parity of qubits 1 and 2)

Critically, both COMMUTE with the logical operators, and:
$S_1|000\rangle = +|000\rangle$, $S_1|111\rangle = +|111\rangle$
so S_1 has the SAME eigenvalue on both logical basis states.

WHY THIS STEP: Because the eigenvalue does not depend on whether the state is |000> or |111>, measuring it reveals NOTHING about a and b.

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Step 5 — The syndrome table

Apply the stabilisers to the corrupted state a|010> + b|101>:

error	$S_1 = Z_0Z_1$	$S_2 = Z_1Z_2$	syndrome
none	+1	+1	00
X on q0	-1	+1	10
X on q1	-1	-1	11
X on q2	+1	-1	01

Every single-qubit error gives a DISTINCT syndrome.

WHY THIS STEP: Four outcomes, four distinguishable cases — the code can locate any single bit flip.

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Step 6 — Why the logical state survives

Take the corrupted state a|010> + b|101>:

S_1 eigenvalue on |010> = -1 ; on |101> = -1 -> SAME
S_2 eigenvalue on |010> = -1 ; on |101> = -1 -> SAME

Both branches share the same syndrome, so measuring it does NOT distinguish them — the superposition a·(…) + b·(…) is preserved intact.

WHY THIS STEP: THIS IS THE HEART OF QEC. The measurement projects onto an error subspace, not onto a logical basis state.

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Step 7 — Correction and generalisation

Syndrome 11 -> apply X to qubit 1:

$X_1 (a|010\rangle + b|101\rangle) = a|000\rangle + b|111\rangle = |\psi\rangle_L$ RESTORED

PHASE FLIPS: conjugate by Hadamards, since H X H = Z.
Encode as |+++> / |---> and the same circuit corrects Z errors.

ARBITRARY ERRORS: any single-qubit error is a linear combination

$E = c_0 I + c_1 X + c_2 Y + c_3 Z$

The syndrome measurement DISCRETISES it — projecting onto one of I, X, Y, Z.

WHY THIS STEP: Continuous errors become discrete ones. This 'digitisation of errors' is what makes fault tolerance possible at all.

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Step 8 — Shor, Steane and the threshold

Shor 9-qubit = phase-flip code (3) concatenated with bit-flip code (3)
-> corrects ANY single-qubit error.
Steane 7-qubit = CSS code from the [7,4] Hamming code — same protection,
fewer qubits, and transversal Clifford gates.
Surface code = 2D lattice, nearest-neighbour only, threshold ~1%.

THRESHOLD THEOREM: if the physical error rate p is below a threshold p_th,
concatenating the code to level L suppresses the logical error rate as
$p_L \approx p_{th} (p/p_{th})^{(2^L)}$ -> doubly exponential suppression.

WHY THIS STEP: Below threshold, adding qubits makes the computer MORE reliable — that is the entire promise of fault-tolerant quantum computing.

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QED

Quantum Error Correction Works — Result

RESULT

The syndrome identifies the error without revealing a or b, so applying the indicated Pauli correction restores the logical state exactly. Below the error threshold, concatenation suppresses errors faster than they accumulate.

WHY IT MATTERS

Without QEC, decoherence destroys any long computation. Shor's algorithm on RSA-2048 needs billions of reliable operations — impossible with raw physical qubits at today's error rates, but achievable with error-corrected logical qubits.

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WHAT YOU'LL DO

Hands-On Activities — Quantum Error Correction

Activity 17

- Bit-Flip and Phase-Flip Error Correction Codes

Activity 18

- Shor 9-Qubit and Steane 7-Qubit Codes

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TOPIC 08 · HANDS-ON ACTIVITY

ACTIVITY 17

Bit-Flip and Phase-Flip Error Correction Codes

Build the three-qubit repetition codes that detect and correct X and Z errors: encode a logical qubit, inject an error, measure the syndrome without collapsing the state, and apply the correction.

MAPS TO LO7 — Synthesize quantum methodologies to address complex problems

YOU'LL PRODUCE

Working bit-flip and phase-flip code circuits that correctly identify and repair an injected single-qubit error.

Tools: Qiskit 2.x, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 17: Bit-Flip and Phase-Flip Error Correction Codes

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ACTIVITY 17 · VERIFY & DEBRIEF

Bit-Flip and Phase-Flip Error Correction Codes

TEST IT — HOW YOU KNOW IT WORKED

The syndrome measurement correctly identifies which qubit carried the injected error without collapsing the logical state, and applying the indicated correction restores the original logical qubit.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC 08 · HANDS-ON ACTIVITY

ACTIVITY 18

Shor 9-Qubit and Steane 7-Qubit Codes

Step up to codes that correct arbitrary single-qubit errors: Shor's 9-qubit concatenated code and Steane's more efficient 7-qubit CSS code, then compare their qubit overhead and discuss the road to fault tolerance.

MAPS TO LO7 — Synthesize quantum methodologies to address complex problems

YOU'LL PRODUCE

Working Shor 9-qubit and Steane 7-qubit code circuits, with a comparison of qubit cost against protection offered.

Tools: Qiskit 2.x, IBM Quantum Runtime, OpenQASM

Full step-by-step guide: Learner Guide — Activity 18: Shor 9-Qubit and Steane 7-Qubit Codes

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ACTIVITY 18 · VERIFY & DEBRIEF

Shor 9-Qubit and Steane 7-Qubit Codes

TEST IT — HOW YOU KNOW IT WORKED

Both the Shor 9-qubit and Steane 7-qubit circuits detect and correct an arbitrary injected single-qubit error (X, Y or Z), and you can state the qubit overhead each code requires.

DISCUSS WITH YOUR TRAINER

What did the result show?

Restate the outcome in your own words.

Why does it work?

Name the quantum principle the activity relies on.

Where would you use it?

Give one realistic application in your own field.

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TOPIC RECAP

Recap — Quantum Error Correction

- Why QEC is needed
- Quantum-specific challenges
- Syndrome measurement
- Bit-flip and phase-flip codes
- Shor, Steane and surface codes
- Fault tolerance

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES ACHIEVED

What You Achieved

1

Quantum foundations

Built the state vectors, unitary and Hermitian operators behind every algorithm.

2

Quantum systems

Visualised qubits on the Bloch sphere and ran circuits on real IBM hardware.

3

Gates and circuits

Applied the full gate set and constructed Bell and GHZ entangled states.

4

Quantum protocols

Implemented teleportation, superdense coding and BB84 with eavesdropper detection.

5

Quantum algorithms

Ran Deutsch, Deutsch-Jozsa, Bernstein-Vazirani, Grover and the QFT.

6

Cryptography and safety

Ran Shor's period finding and built a PQC migration plan.

7

Applications

Simulated H_2 with VQE and optimised with QAOA.

8

Error correction

Encoded, diagnosed and corrected errors with the bit-flip, Shor and Steane codes.

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NEXT STEPS

Continue Your Quantum Journey

1. **IBM Quantum Learning**
   Free courses and challenges at learning.quantum.ibm.com.

2. **Qiskit documentation**
   The reference for every gate, primitive and runtime feature.

3. **PennyLane demos**
   Excellent tutorials on variational and quantum-machine-learning methods.

4. **NIST PQC**
   Track FIPS 203/204/205 adoption and vendor migration guidance.

5. **Practise the labs**
   Re-run every activity in this course from a blank notebook.

6. **Stay current**
   Quantum hardware roadmaps shift yearly — follow the vendor announcements.

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FINAL ASSESSMENT

Assessment

Written Assessment (WA)

- Short-Answer Questions (SAQ)
- 1 hour · open book
- Tests underpinning KNOWLEDGE
- Covers all eight topics

Practical Performance (PP)

- Hands-on quantum tasks
- 1 hour · open book
- Tests PRACTICAL ability
- Built on the class activities

OPEN BOOK — course slides, Learner Guide and approved materials only

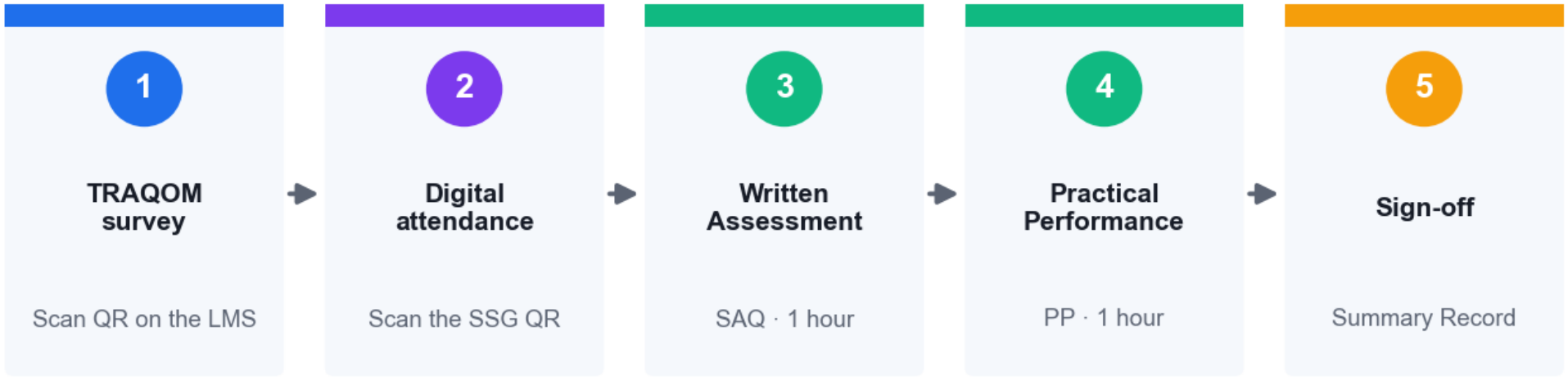
Eligibility: minimum 75% attendance + assessed Competent. An appeal process is available.

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ON ASSESSMENT DAY

Assessment Flow

What happens on assessment day



Open book — course slides, Learner Guide and approved materials only

Complete every step in order — the TRAQOM survey and digital attendance are mandatory for funding.

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Digital Attendance (Mandatory)

1. Trainer displays the digital attendance QR code from the SSG portal
2. Scan the QR code with your mobile phone camera
3. Submit your attendance — AM, PM and Assessment, every day
4. 75% minimum attendance is required for assessment and funding

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END OF COURSE

Thank You!

You can now design quantum models, apply quantum gates, select quantum algorithms and plan a post-quantum migration — the skills that secure the future.

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